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The equation $x^4 + x + y^3 + 8y + 1 = 0$ has no integer solutions

Size $h = 43$, length $l = 12$

Theorem 1. *The equation*

$$x^4 + x + y^3 + 8y + 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses 46 auxiliary polynomials $H_0, \dots, H_{45}$, 5 of which are constants, and 2 graphs on these vertices with 60 and 86 edges, respectively; to each edge $\{i, j\}$ we attach the Hilbert symbol $(H_i, H_j)_v$ of the values at a hypothetical solution. By the product formula, the product of these symbols over all edges and all places is 1. We show that the product over the edges is 1 at every odd prime outside a finite set S , compute it at the primes of S and at the real place, and obtain a contradiction. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. *Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.*

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

Lemma 4. *Let p be a prime. For a non-zero integer X , write $X = p^j u$ with $p \nmid u$. If p is odd, put $c_p(X) = (j, e) \in \mathbb{F}_2^2$, where $(u/p) = (-1)^e$; if $p = 2$, put $c_2(X) = (j, e_1, e_2) \in \mathbb{F}_2^3$, where $u \equiv (-1)^{e_1} 5^{e_2} \pmod{8}$; all entries are taken modulo 2. Then $c_p(XX') = c_p(X) + c_p(X')$ and $(X, X')_p = (-1)^{\beta_p(c_p(X), c_p(X'))}$ for non-zero integers X, X' , where β_p is the symmetric bilinear form*

$$\begin{aligned} \beta_p((j, e), (j', e')) &= jj' \varepsilon(p) + je' + j'e && \text{if } p \text{ is odd,} \\ \beta_2((j, e_1, e_2), (j', e'_1, e'_2)) &= e_1 e'_1 + je'_2 + j'e_2. \end{aligned}$$

Proof. For odd p , this is Theorem 2(ii), as the Legendre symbol is multiplicative. For $p = 2$, the odd residues 1, 3, 5, 7 modulo 8 are $(-1)^{e_1} 5^{e_2}$ with $(e_1, e_2) = (0, 0), (1, 1), (0, 1), (1, 0)$, respectively; in each case $\varepsilon(u) \equiv e_1$ and $\omega(u) \equiv e_2 \pmod{2}$, so Theorem 2(iii) gives the formula, and c_2 is additive because $(e_1, e_2) \mapsto (-1)^{e_1} 5^{e_2}$ is an isomorphism from $\mathbb{F}_2^2$ to the group of odd residues modulo 8. $\square$

Lemma 5. *Let p be a prime, $k \geq 1$, and let X be a non-zero integer with $X \equiv r \pmod{p^k}$, where $0 \leq r < p^k$. Then $c_p(X) \in b + D$ for the vector b and the subspace D given as follows. If $r = 0$, then $b = 0$ and D is the whole space. Otherwise write $r = p^j r'$ with $p \nmid r'$, so that $j < k$. If p is odd, or if $p = 2$ and $k - j \geq 3$, then $b = c_p(r)$ and $D = 0$. If $p = 2$ and $k - j = 2$, then $b = (j, e_1, 0)$ with $r' \equiv (-1)^{e_1} \pmod{4}$, and D is spanned by $(0, 0, 1)$. If $p = 2$ and $k - j = 1$, then $b = (j, 0, 0)$, and D is spanned by $(0, 1, 0)$ and $(0, 0, 1)$.*

Proof. If $r \neq 0$, then $v_p(X) = j$ and $X/p^j \equiv r' \pmod{p^{k-j}}$. For odd p , the Legendre symbol of X/p^j is therefore that of r' . For $p = 2$, the residue of $X/2^j$ modulo 8 is that of r' if $k - j \geq 3$, and its residue modulo 4, which determines e_1 , is that of r' if $k - j = 2$. $\square$

Lemma 6. *Let p be a prime, and let E be a set of edges on a finite set of vertices. For every vertex i , let b_i be a vector and D_i a subspace of $\mathbb{F}_2^2$ (of $\mathbb{F}_2^3$ if $p = 2$), and put $\sigma_i = \sum_{j \sim i} b_j$, the sum over the edges $\{i, j\} \in E$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every vertex i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge $\{i, j\} \in E$ and all $d \in D_i$ and $d' \in D_j$. Then*

$$\sum_{\{i, j\} \in E} \beta_p(c_i, c_j) = \sum_{\{i, j\} \in E} \beta_p(b_i, b_j) \quad \text{whenever } c_i \in b_i + D_i \text{ for every } i.$$

By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and symmetric, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{\{i, j\} \in E} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

Lemma 7. *Let $f(z) = az^3 + bz^2 + cz + d$ be a real polynomial with $a \neq 0$ and discriminant $\Delta < 0$. Then f has exactly one real root η , and $f(z)$ has the sign of $a(z - \eta)$ for every real $z \neq \eta$.*

Proof. The discriminant is $\Delta = a^4 \prod_{i < j} (\rho_i - \rho_j)^2$, where ρ_1, ρ_2, ρ_3 are the complex roots of f . If all roots were real, then $\Delta \geq 0$. Hence f has a non-real root ρ ; then $\bar{\rho} \neq \rho$ is also a root, and the third root η is real. Thus $f(z) = a(z - \eta)(z - \rho)(z - \bar{\rho}) = a(z - \eta)|z - \rho|^2$ for real z , and $|z - \rho|^2 > 0$. $\square$

2 The graph

Put $F(x, y) = x^4 + x + y^3 + 8y + 1$, so that (1) is the equation $F(x, y) = 0$. The vertices are the polynomials

$$\begin{aligned} H_0 &= -y, \\ H_1 &= -x - y, \\ H_2 &= -y - 1, \\ H_3 &= -y + 1, \\ H_4 &= x - y, \\ H_5 &= -2x - y, \\ H_6 &= -x - y + 1, \\ H_7 &= x - y + 1, \\ H_8 &= 2x - y, \\ H_9 &= -2x - y + 1, \\ H_{10} &= -x - y - 2, \\ H_{11} &= 2x - y - 1, \\ H_{12} &= x^2 + y - 1, \\ H_{13} &= x^2 - y + 1, \\ H_{14} &= x^2 + x - y, \\ H_{15} &= x^2 - x + y - 1, \\ H_{16} &= x^2 - x + y + 1, \\ H_{17} &= x^2 + y - 2, \\ H_{18} &= x^2 + x + y + 1, \\ H_{19} &= x^2 + 2x + y, \\ H_{20} &= x^2 - 2x - y, \\ H_{21} &= x^2 - x - y - 1, \\ H_{22} &= x^2 - x - y + 1, \\ H_{23} &= x^2 + x - y - 1, \\ H_{24} &= x^2 + 2x - y, \\ H_{25} &= x^2 + x + y - 2, \\ H_{26} &= x^2 + 2x + y - 1, \\ H_{27} &= -1, \\ H_{28} &= 1, \\ H_{29} &= 2, \\ H_{30} &= -3, \\ H_{31} &= -x - y - 1, \\ H_{32} &= -y - 2, \\ H_{33} &= -y + 2, \\ H_{34} &= x - y - 1, \\ H_{35} &= -x - y + 2, \\ H_{36} &= 2x - y + 1, \\ H_{37} &= -2x - y + 2, \end{aligned}$$

$$\begin{aligned}
H_{38} &= 2x - y - 2, \\
H_{39} &= 2x - y + 2, \\
H_{40} &= x^2 + y, \\
H_{41} &= x^2 - y - 2, \\
H_{42} &= x^2 - 2x + y + 1, \\
H_{43} &= x^2 - x + y - 2, \\
H_{44} &= x^2 - x + y + 2, \\
H_{45} &= 3.
\end{aligned}$$

There are 2 graphs on these vertices. The edges of the first graph are $\{0, 3\}, \{0, 4\}, \{0, 11\}, \{0, 13\}, \{0, 14\}, \{0, 16\}, \{0, 25\}, \{1, 5\}, \{1, 9\}, \{1, 20\}, \{2, 20\}, \{3, 8\}, \{3, 16\}, \{3, 18\}, \{4, 13\}, \{5, 8\}, \{6, 18\}, \{6, 25\}, \{7, 10\}, \{7, 13\}, \{7, 14\}, \{7, 18\}, \{7, 22\}, \{7, 23\}, \{9, 12\}, \{9, 15\}, \{9, 21\}, \{9, 23\}, \{10, 11\}, \{10, 13\}, \{10, 15\}, \{10, 19\}, \{11, 16\}, \{11, 18\}, \{11, 22\}, \{11, 23\}, \{12, 13\}, \{12, 16\}, \{12, 17\}, \{12, 26\}, \{13, 14\}, \{13, 21\}, \{13, 26\}, \{14, 24\}, \{15, 22\}, \{15, 23\}, \{18, 19\}, \{19, 25\}, \{20, 21\}, \{20, 24\}, \{25, 26\}, \{0, 27\}, \{4, 29\}, \{6, 30\}, \{10, 29\}, \{14, 27\}, \{16, 27\}, \{18, 29\}, \{22, 29\}, \{23, 27\}$. The edges of the second graph are $\{0, 6\}, \{0, 9\}, \{0, 10\}, \{0, 36\}, \{0, 13\}, \{1, 7\}, \{1, 37\}, \{1, 14\}, \{1, 18\}, \{1, 41\}, \{2, 32\}, \{2, 9\}, \{2, 11\}, \{2, 40\}, \{2, 44\}, \{3, 8\}, \{3, 9\}, \{3, 11\}, \{3, 15\}, \{3, 16\}, \{3, 18\}, \{4, 10\}, \{4, 40\}, \{4, 13\}, \{5, 6\}, \{5, 33\}, \{31, 11\}, \{31, 36\}, \{31, 16\}, \{31, 21\}, \{31, 42\}, \{6, 38\}, \{6, 40\}, \{33, 37\}, \{33, 13\}, \{33, 43\}, \{34, 23\}, \{7, 40\}, \{7, 44\}, \{8, 14\}, \{8, 16\}, \{9, 11\}, \{9, 15\}, \{9, 16\}, \{9, 21\}, \{9, 43\}, \{10, 37\}, \{10, 13\}, \{10, 15\}, \{35, 37\}, \{35, 40\}, \{35, 18\}, \{11, 36\}, \{11, 40\}, \{11, 16\}, \{11, 23\}, \{11, 44\}, \{36, 38\}, \{36, 16\}, \{36, 43\}, \{37, 38\}, \{37, 39\}, \{37, 18\}, \{39, 44\}, \{40, 43\}, \{13, 14\}, \{13, 21\}, \{14, 41\}, \{15, 23\}, \{16, 42\}, \{18, 23\}, \{21, 41\}, \{42, 43\}, \{1, 30\}, \{2, 29\}, \{5, 29\}, \{6, 29\}, \{7, 29\}, \{8, 45\}, \{10, 27\}, \{11, 45\}, \{37, 29\}, \{15, 29\}, \{16, 27\}, \{43, 27\}, \{44, 29\}$. For a vertex i , we write $j \sim_g i$ if $\{i, j\}$ is an edge of the g -th graph. For every non-constant vertex i and every graph g in which i has a neighbour, Table 1 gives a positive integer $m_{g,i}$ and polynomials $W_{g,i}$, $A_{g,i}$ and $B_{g,i}$ with integer coefficients such that

$$m_{g,i}^2 \prod_{j \sim_g i} H_j - W_{g,i}^2 = A_{g,i}F + B_{g,i}H_i. \quad (2)$$

For every edge $\{i, j\}$ between two non-constant vertices, Table 2 gives a positive integer k_{ij} and polynomials α_{ij} , β_{ij} and γ_{ij} with integer coefficients such that

$$\alpha_{ij}F + \beta_{ij}H_i + \gamma_{ij}H_j = k_{ij}. \quad (3)$$

These identities can be checked by expanding both sides. For the signs of the vertices, we regard F as a polynomial in y ; it has degree 3 and leading coefficient 1, and its discriminant is

$$\Delta(x) = -27x^8 - 54x^5 - 54x^4 - 27x^2 - 54x - 2075.$$

For $i \in I = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44\}$, the vertex H_i has the form $k_i y + r_i(x)$ with a non-zero integer k_i , and Table 3 lists the polynomial $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$, which has integer coefficients.

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) .

Step 1: the values are non-zero. The non-constant H_i are non-zero by Step 2.

Step 2: signs. *The signs of H_i for $i \in I$.* If $\Delta(x) < 0$, then by Lemma 7, applied to $F(x, z)$ as a polynomial in z , y is its only real root, and $F(x, z)$ has the sign of $(z - y)$ for $z \neq y$. For $i \in I$ and $z = -r_i(x)/k_i$, we have $z - y = -H_i/k_i$, so if moreover $\varphi_i(x) = k_i^3 F(x, z) \neq 0$, then $H_i \neq 0$, and H_i has the sign of $-\varphi_i(x)$. Expanding $\Delta(9 + t)$ and the $\varphi_i(9 + t)$ as polynomials in t gives in each case coefficients of one sign and a non-zero constant term, and likewise for $\Delta(-11 - t)$ and the $\varphi_i(-11 - t)$. Hence, for $x \geq 9$ and for $x \leq -11$, we have $\Delta(x) < 0$, and $\varphi_i(x)$ is non-zero and has the sign of the constant term of the corresponding expansion; the resulting signs of the H_i are listed in Table 3. It remains to consider $-10 \leq x \leq 8$. Checking the residues of y , we find that for $x \in \{-10, -9, -7, -6, -4, -3, -1, 0, 2, 3, 5, 6, 8\}$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , for $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , for $x \in \{-8, -2, 7\}$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , for $x = -5$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , and for $x = 4$, we have $F(x, y) \not\equiv 0 \pmod{37}$ for every y . Hence (1) has no solution with $-10 \leq x \leq 8$.

For a place v , put $B_v^{(g)} = \prod_{\{i,j\}} (H_i, H_j)_v$, the product over the edges of the g -th graph. By Theorem 2(i), $(H_i, H_j)_\infty = -1$ exactly when $H_i < 0$ and $H_j < 0$. Here $H_i > 0$ for $i \in \{28, 29, 45\}$ and $H_i < 0$ for $i \in \{27, 30\}$. The other signs are given in Table 3 for $x \geq 9$ and for $x \leq -11$. In both cases, counting the edges of each graph whose two values are negative gives $(B_\infty^{(g)})_g = (1, 1)$.

Step 3: odd primes outside S . Let $S = \{2, 3, 5, 7, 11, 13\}$ be the set of primes dividing 2, the $m_{g,i}$, the k_{ij} and the constant vertices. Let $p \notin S$ be an odd prime; we show that $B_p^{(g)} = 1$ for every g . For an edge $\{i, j\}$ of the g -th graph such that p divides neither H_i nor H_j , $(H_i, H_j)_p = 1$ by Theorem 2(ii). Now let i be a vertex with $p \mid H_i$. Then H_i is not constant, as $p \notin S$. For every edge $\{i, j\}$ of the g -th graph, $p \nmid H_j$: if H_j is constant, because $p \notin S$, and otherwise by (3), because $p \nmid k_{ij}$. In particular, no edge has both values divisible by p . By Theorem 2(v), the edges at i contribute $\prod_{j \sim_g i} (H_i, H_j)_p = (H_i, \Pi_i)_p$, where $\Pi_i = \prod_{j \sim_g i} H_j$ is not divisible by p . By (2), $m_{g,i}^2 \Pi_i \equiv W_{g,i}^2 \pmod{p}$, and $p \nmid m_{g,i}$, so Π_i is a non-zero square modulo p , and $(H_i, \Pi_i)_p = (\Pi_i, H_i)_p = 1$ by Lemma 3. Hence $B_p^{(g)} = 1$.

Step 4: the primes in S . Let $p \in S$. By Lemma 4, $B_p^{(g)} = (-1)^{N_g}$, where $N_g = \sum_{\{i,j\}} \beta_p(c_p(H_i), c_p(H_j))$, the sum over the edges of the g -th graph. Suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Then $H_i \equiv H_i(a, b) \pmod{p^k}$, and Lemma 5 gives a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$; for a constant vertex, we take $b_i = c_p(H_i)$ and $D_i = 0$. If the hypotheses of Lemma 6 hold for the edges of every graph, then $N_g = \sum_{\{i,j\}} \beta_p(b_i, b_j)$, so $B_p^{(g)}$ is determined by a , b and k . Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a$, $y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until these hypotheses hold, or until no solution remains. Table 4 lists the resulting terminal classes together with the value of the vector $(B_p^{(g)})_g$; every integer solution lies in one of them. The result is $(B_2^{(g)})_g \in \{(1, 1)\}$, $(B_3^{(g)})_g \in \{(1, -1), (-1, 1), (-1, -1)\}$, $(B_5^{(g)})_g \in \{(1, 1)\}$, $(B_7^{(g)})_g \in \{(1, 1)\}$, $(B_{11}^{(g)})_g \in \{(1, 1)\}$, $(B_{13}^{(g)})_g \in \{(1, 1)\}$.

Step 5: contradiction. By Theorem 2(iv), applied to each edge, $\prod_v B_v^{(g)} = 1$ for every g . By Step 3, the factors with $p \notin S$ are 1, so the componentwise product of the vectors $(B_p^{(g)})_g$ with $p \in S$ is $(B_\infty^{(g)})_g = (1, 1)$ by Step 2. But by Step 4, this product is $(1, -1)$, $(-1, 1)$, or $(-1, -1)$. This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.

- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The data of the identities (2).

$g = 1, i = 0, m = 1$, neighbours 27, 3, 4, 11, 13, 14, 16, 25	
W	$x^3 - 2x^2 + x + 3$
A	$2x^6y - 2x^6 - 3x^5y^2 + 4x^5y - x^5 + x^4y^3 - 3x^4y^2 - x^4y + 3x^4$ $+ x^3y^3 + 5x^3y^2 - 3x^3y - 3x^3 - 2x^2y^4 - 6x^2y^3 - 5x^2y^2$ $+ 6x^2y + 6x^2 + 3xy^5 + 3xy^4 + 8xy^3 - 22xy^2 + 9xy + 3x - y^6 + y^5 + 25y^3$ $+ 4y^2 - 26y - 9$
B	$x^3y^5 + 6x^3y^4 + 5x^3y^3 + 28x^3y^2 - 13x^3y - 26x^3 - 2x^2y^6$ $- 6x^2y^5 - 22x^2y^4 - 20x^2y^3 - 51x^2y^2 + 10x^2y + 74x^2 + 3xy^7$ $+ 3xy^6 + 35xy^5 - 7xy^4 + 77xy^3 - 123xy^2 + 49xy + 3x - y^8 + y^7 - 9y^6 + 34y^5$ $+ 7y^4 + 170y^3 + 47y^2 - 202y - 98$
$g = 1, i = 1, m = 1$, neighbours 5, 9, 20	
W	$-x^2 + x$
A	0
B	$-3x^3 - x^2y + 8x^2 + 5xy - 3x + y^2 - y$
$g = 1, i = 2, m = 1$, neighbours 20	
W	$-x + 1$
A	0
B	1
$g = 1, i = 3, m = 1$, neighbours 0, 8, 16, 18	
W	$-x^2 + 3x + 2$
A	$-2xy + y^2 - 1$
B	$4x^3y + 6x^3 - 2x^2y^2 - 3x^2y - 5x^2 - 2xy^3 - 11xy - 11x + y^4 + 5y^2 + 5y - 3$
$g = 1, i = 4, m = 2$, neighbours 29, 0, 13	
W	$-x^3 - 1$
A	$-x^2 + y - 1$
B	$-x^3y + x^3 - x^2y^2 + x^2y - x^2 + xy^2 - xy + x + y^3 - y^2 + 1$
$g = 1, i = 5, m = 1$, neighbours 1, 8	
W	$2x$
A	0
B	$3x - y$
$g = 1, i = 6, m = 1$, neighbours 30, 18, 25	
W	$x^3 - 4$
A	$-x^2 + y - 2$
B	$x^3y + x^3 - x^2y^2 - 2x^2 - xy^2 - 3x + y^3 - y^2 + 10y - 8$
$g = 1, i = 7, m = 1$, neighbours 10, 13, 14, 18, 22, 23	
W	$6x^3 + 37x^2 + 62x + 30$
A	$-x^7 - x^6y - 4x^6 + x^5y - 6x^5 + 3x^4y^2 + 10x^4y - 6x^4 + x^3y^3$ $+ 4x^3y^2 + 26x^3y - 3x^3 + x^2y^4 + 2x^2y^3 + 6x^2y^2 + 53x^2y - 32x^2$ $- xy^4 - 2xy^3 - 26xy^2 + 54xy - 437x - 5y^5 - 15y^4 - 44y^3 - 15y^2 + 395y - 406$
B	$2x^3y^5 + 3x^3y^4 + 13x^3y^3 - 90x^3y^2 - 369x^3y - 1700x^3 + x^2y^6$ $- 3x^2y^5 - 20x^2y^4 - 130x^2y^3 - 503x^2y^2 - 1378x^2y - 3213x^2$ $- 6xy^6 - 28xy^5 - 169xy^4 - 387xy^3 - 1292xy^2 - 1689xy - 2381x - 5y^7 - 20y^6$ $- 103y^5 - 241y^4 - 215y^3 - 789y^2 + 2357y - 494$
$g = 1, i = 8, m = 3$, neighbours 3, 5	
W	$-2x^2 - 8x - 2$
A	-4
B	$-16x^2 - 8xy - 36x - 4y^2 - 9y - 23$
$g = 1, i = 9, m = 1$, neighbours 1, 12, 15, 21, 23	
W	$-13x^3 + 8x^2 - 17x + 10$
A	$-x^5 - x^4y + x^4 + x^3y + 5x^3 + 4x^2y - 172x^2 + xy^3 + xy^2 + 7xy + 200x + y^4$ $- 5y^3 + 55y^2 - y - 60$
B	$-3x^3y^3 + 23x^3y^2 + 7x^3y + 224x^3 + 2x^2y^4 - 10x^2y^3 + 13x^2y^2$ $- 87x^2y - 240x^2 - xy^5 + 8xy^4 - 96xy^3 + 67xy^2 - 608xy + 119x + y^6 - 4y^5$ $+ 60y^4 + 20y^3 + 393y^2 + 440y - 40$
$g = 1, i = 10, m = 1$, neighbours 29, 7, 11, 13, 15, 19	
W	$-6x^3 - 20x^2 - 18x - 4$
A	$4x^4 - 6x^3y + 6x^3 + 2x^2y^2 - 2x^2y - 44x^2 - 4xy^2 + 14xy - 254x - 10y^3$ $+ 18y^2 - 96y - 348$
B	$-8x^3y^3 + 26x^3y^2 - 90x^3y + 288x^3 + 2x^2y^4 - 10x^2y^3$

$$\begin{aligned}
& +24x^2y^2 - 106x^2y + 166x^2 + 6xy^4 - 42xy^3 + 56xy^2 - 300xy - 148x - 10y^5 \\
& +40y^4 - 260y^3 + 306y^2 - 1358y - 166 \\
g = 1, i = 11, m = 1, \text{ neighbours } 0, 10, 16, 18, 22, 23 \\
W & -181x^3 + 259x^2 - 426x + 144 \\
A & x^5y + x^4y^2 + 2x^4y + x^2y - 32761x^2 - xy^4 - 2xy^3 - 9xy^2 + 93758x - y^5 \\
& +20y^4 - 406y^3 + 8198y^2 - 33118y - 39151 \\
B & -12x^3y^4 + 196x^3y^3 - 4108x^3y^2 + 16558x^3y - 91071x^3 - 6x^2y^5 \\
& +92x^2y^4 - 1956x^2y^3 + 6227x^2y^2 - 37256x^2y + 107243x^2 - 3xy^6 \\
& +43xy^5 - 934xy^4 + 18511xy^3 - 15519xy^2 + 166036xy - 104911x - y^7 + 21y^6 \\
& -436y^5 + 8789y^4 - 45139y^3 + 71168y^2 - 327909y - 18415 \\
g = 1, i = 12, m = 1, \text{ neighbours } 9, 13, 16, 17, 26 \\
W & 2x^3 - 4x^2 \\
A & -2x^5 - x^4y - x^4 + 2x^3y^2 - x^3y + 17x^3 - 2x^2y^2 + 3x^2y - 9x^2 \\
& +2xy^3 - xy^2 + 14xy - 15x - y^3 + 4y^2 - 13y + 10 \\
B & -2x^5y^2 - 4x^5y - 10x^5 + 2x^3y^3 + 2x^3y^2 + 6x^3y + 28x^3 + x^2y^4 \\
& +2x^2y^3 + 3x^2y^2 + 21x^2y - 33x^2 - 2xy^5 - xy^4 - 28xy^3 - 13xy^2 - 133xy \\
& +5x + y^5 - 2y^4 + 15y^3 - 22y^2 + 80y + 8 \\
g = 1, i = 13, m = 1, \text{ neighbours } 0, 4, 7, 10, 12, 14, 21, 26 \\
W & -8x^4 - 2x^3 - 50x^2 - 6x - 60 \\
A & x^7y - x^6y^2 + 5x^6y - x^5y^3 - 6x^5y^2 + 4x^5y + x^4y^4 - x^4y^3 \\
& -8x^4y^2 - 16x^4y - 64x^4 - x^3y^5 - 6x^3y^4 - 8x^3y^3 - 64x^3y^2 \\
& -3x^3y - 102x^3 + x^2y^5 - 5x^2y^4 + 13x^2y^3 - 7x^2y^2 - 8x^2y \\
& -804x^2 + xy^6 + 6xy^5 + 5xy^4 + 40xy^3 - 26xy^2 + 36xy - 98x - 2y^6 + 10y^5 \\
& -25y^4 + 67y^3 - 15y^2 + 497y - 892 \\
B & x^5y^5 + 7x^5y^4 + 14x^5y^3 + 69x^5y^2 - 29x^5y + 70x^5 + x^3y^6 \\
& +6x^3y^5 + 7x^3y^4 + 55x^3y^3 - 98x^3y^2 + 99x^3y - 204x^3 - x^2y^7 \\
& +x^2y^6 - 10x^2y^5 + 68x^2y^4 + 72x^2y^3 + 179x^2y^2 + 45x^2y \\
& -2426x^2 + xy^8 + 7xy^7 + 19xy^6 + 108xy^5 + 136xy^4 + 470xy^3 + 278xy^2 + 521xy \\
& +270x - 2y^8 + 7y^7 - 34y^6 + 115y^5 - 92y^4 + 913y^3 - 30y^2 + 3931y - 2708 \\
g = 1, i = 14, m = 1, \text{ neighbours } 27, 0, 7, 13, 24 \\
W & x^3 - x \\
A & -x^3 - x^2 + xy \\
B & x^5 - x^4 + x^3y + x^3 - x^2y^2 + x^2y + 2x^2 + xy^3 - 2xy^2 + 11xy + y^3 \\
& -2y^2 + y \\
g = 1, i = 15, m = 1, \text{ neighbours } 9, 10, 22, 23 \\
W & 2x^4 - 6x^3 - 2x^2 + 6x \\
A & -4x^4 + 11x^3 - 26x^2 + 3xy + 16x + 4y^2 - 27y + 23 \\
B & 13x^5 + 13x^4 - 13x^3y - 31x^3 + 4x^2y^3 - 3x^2y^2 + 30x^2y + 16x^2 \\
& -7xy^3 + 4xy^2 - 49xy + 25x - 4y^4 + 24y^3 - 30y^2 + 179y + 21 \\
g = 1, i = 16, m = 2, \text{ neighbours } 27, 0, 3, 11, 12 \\
W & x^5 - 3x^4 + 5x^3 - 9x^2 + 8x - 4 \\
A & -x^6 + 6x^5 - 19x^4 - 5x^3y^2 + 3x^3y - 11x^3 + x^2y^3 + 8x^2y - 100x^2 \\
& -6xy^3 - 48xy + 84x + y^4 - 5y^3 + 14y^2 - 98y - 116 \\
B & 5x^5y^2 - 3x^5y + 60x^5 + 5x^4y^2 - 3x^4y + 60x^4 - 5x^3y^3 + 3x^3y^2 \\
& -60x^3y + 75x^3 - x^2y^4 + 14x^2y^3 - 8x^2y^2 + 130x^2y - 24x^2 + 5xy^5 \\
& +xy^4 + 66xy^3 + 22xy^2 + 200xy + 196x - y^6 + 6y^5 - 28y^4 + 169y^3 - 164y^2 + 930y \\
& +100 \\
g = 1, i = 17, m = 1, \text{ neighbours } 12 \\
W & 1 \\
A & 0 \\
B & 1 \\
g = 1, i = 18, m = 1, \text{ neighbours } 29, 3, 6, 7, 11, 19 \\
W & x^5 + 4x^4 + 5x^3 + 8x^2 + 6x \\
A & -x^6 - 8x^5 - 26x^4 + 7x^3y^2 - 12x^3y + 31x^3 + x^2y^3 + 8x^2y - 92x^2 \\
& +8xy^3 - 7xy^2 + 80xy - 69x + y^4 - 7y^3 + 24y^2 - 97y - 129 \\
B & -7x^5y^2 + 12x^5y - 86x^5 + 7x^4y^2 - 12x^4y + 86x^4 + 7x^3y^3 \\
& -5x^3y^2 + 74x^3y - 29x^3 - x^2y^4 + 19x^2y^3 - 23x^2y^2 + 177x^2y \\
& -63x^2 - 7xy^5 + 6xy^4 - 113xy^3 + 55xy^2 - 482xy + 65x - y^6 + 8y^5 - 38y^4 \\
& +186y^3 - 242y^2 + 998y + 129 \\
g = 1, i = 19, m = 1, \text{ neighbours } 10, 18, 25
\end{aligned}$$

W	$x^2 + x - 2$
A	0
B	$-x^3 - x^2y - 3x^2 - xy - y^2 - y + 4$
$g = 1, i = 20, m = 1, \text{neighbours } 1, 2, 21, 24$	
W	$2x^3 - 4x^2 + 2x$
A	$-x^3 - 4x^2 + xy + 12x + 6y$
B	$x^5 + 2x^4 + x^3y + 9x^3 + x^2y^2 - 4x^2 + xy^3 + 2xy^2 + 13xy + 10x + 5y^3$ $-2y^2 + 47y + 6$
$g = 1, i = 21, m = 1, \text{neighbours } 9, 13, 20$	
W	$x^2 + x - 2$
A	0
B	$-2x^3 - x^2y + 2x^2 + 3xy - 6x + y^2 - 3y + 4$
$g = 1, i = 22, m = 1, \text{neighbours } 29, 7, 11, 15$	
W	$2x^3 - 6x^2 + 4x$
A	$-4x^2 + 4x + 4y - 4$
B	$20x^3 - 4x^2y - 24x^2 + 10xy + 6x + 4y^3 - 2y^2 + 32y + 6$
$g = 1, i = 23, m = 1, \text{neighbours } 27, 7, 9, 11, 15$	
W	$x^5 + x^4 - x^3 - x^2$
A	$-x^6 - 2x^5 + x^4 - x^3y^2 - 3x^3 + x^2y^3 + 8x^2y + 4x^2 + 2xy^3 + xy^2$ $+16xy + 18x - y^4 - y^3 - 9y^2 - 21y - 12$
B	$x^5y^2 + 8x^5 - x^4y^2 - 8x^4 + x^3y^3 + x^3y^2 + 8x^3y + x^3$ $+x^2y^4 - 2x^2y^3 + 8x^2y^2 - 7x^2y + xy^5 + 14xy^3 + xy^2 + 44xy - 7x - y^6$ $-17y^4 - 14y^3 - 69y^2 - 108y - 11$
$g = 1, i = 24, m = 1, \text{neighbours } 14, 20$	
W	$2x$
A	0
B	$x^2 - 3x - y$
$g = 1, i = 25, m = 1, \text{neighbours } 0, 6, 19, 26$	
W	$x^3 + 2x^2 - x - 2$
A	$x^3 - x^2 + xy - 4x - 2y + 4$
B	$-x^5 + x^4 + x^3y - 3x^3 + x^2y^2 + 3x^2y - 2x^2 - xy^3 + 4xy^2 - 7xy + 4x$ $+3y^3 + 17y + 4$
$g = 1, i = 26, m = 1, \text{neighbours } 12, 13, 25$	
W	$-2x^2 - 2x$
A	$x^2 + 2x + y - 1$
B	$-x^3 - 3x^2 + xy - 4x - y^3 - y^2 - 5y - 3$
$g = 2, i = 0, m = 1, \text{neighbours } 6, 9, 10, 36, 13$	
W	$x^3 + 4x + 1$
A	$-5x^2 - 8xy - 4x - 3y^2 - 2y - 3$
B	$-2x^3y^2 - 5x^3y + 4x^3 - x^2y^3 - 7x^2y^2 - 38x^2 - 6xy^3 - 9xy^2 - 64xy$ $-41x - 2y^4 - 4y^3 - 29y^2 - 11y - 33$
$g = 2, i = 1, m = 1, \text{neighbours } 30, 7, 37, 14, 18, 41$	
W	$-3x^3 + 6x^2 - 15x$
A	$6x^4 - 3x^3y + 12x^3 - 3x^2y^2 - 3x^2y - 15x^2 - 3xy^2 + 6xy - 3y^3 - 24y$
B	$9x^3y^2 + 24x^3y + 168x^3 - 3x^2y^4 - 6x^2y^3 - 51x^2y^2 - 69x^2y$ $-189x^2 + 6xy^3 + 27xy^2 + 72xy + 186x - 3y^5 + 3y^4 - 48y^3 - 18y^2 - 192y - 12$
$g = 2, i = 2, m = 1, \text{neighbours } 29, 32, 9, 11, 40, 44$	
W	$-8x + 8$
A	$8x^2y + 16x^2 - 16xy - 32x - 2y^3 + 12y^2 + 58y + 52$
B	$-2x^3y^2 + 22x^3y + 52x^3 + 12x^2y^3 + 8x^2y^2 + 20x^2y + 40x^2 - 18xy^3$ $-12xy^2 - 70xy - 108x - 2y^5 + 16y^4 + 34y^3 + 118y^2 + 350y + 116$
$g = 2, i = 3, m = 1, \text{neighbours } 8, 9, 11, 15, 16, 18$	
W	$8x^3 - 2x^2 - 2x + 20$
A	$-8x^5 + 4x^4y + 16x^4 + 2x^3y^2 - 32x^3y - 10x^3 - x^2y^3 + 10x^2y^2$ $+45x^2y - 46x^2 + 15xy^3 - 44xy^2 + 25xy + 16x - 7y^4 - 8y^3 + 11y^2 - 68y + 32$
B	$2x^3y^4 - 38x^3y^3 - 12x^3y^2 - 220x^3y - 272x^3 - x^2y^5 + 12x^2y^4$ $+48x^2y^3 + 81x^2y^2 + 394x^2y + 100x^2 + 15xy^5 - 32xy^4 + 106xy^3$ $-215xy^2 - 46xy + 34x - 7y^6 - 14y^5 - 58y^4 - 199y^3 - 89y^2 - 621y - 432$
$g = 2, i = 4, m = 1, \text{neighbours } 10, 40, 13$	

W	$-4x$
A	$-x - y$
B	$-2x^3 - 2x^2y - x^2 - 2xy^2 - 2xy - 17x - y^3 - y^2 - 9y + 1$
$g = 2, i = 5, m = 1$, neighbours 29, 6, 33	
W	$2x + 2$
A	0
B	$2x - 2y + 6$
$g = 2, i = 6, m = 1$, neighbours 29, 0, 5, 38, 40	
W	$-x^3 + 3x^2 - 3x - 2$
A	$-x^2 + 6x + 7y - 8$
B	$-x^3y + 7x^3 + x^2y^2 - 8x^2 - 3xy^2 - 6x + 9y^3 + 5y^2 + 61y + 4$
$g = 2, i = 7, m = 1$, neighbours 29, 1, 40, 44	
W	$-x^3 + x^2 + x + 2$
A	$-x^2 - y$
B	$-x^3y + 3x^3 - x^2y^2 + 2x^2y - 4x^2 - xy^2 - y^3 + y^2 - 3y - 4$
$g = 2, i = 8, m = 3$, neighbours 45, 3, 14, 16	
W	$-2x^3 + 6x^2 + 6x - 1$
A	$-4x^2 + 24x - 11y - 1$
B	$-8x^3y + 8x^3 - 4x^2y^2 + 4x^2y - 36x^2 + 2xy^2 - 2xy - 22x - 11y^3 - 28y^2 - 88y + 8$
$g = 2, i = 9, m = 1$, neighbours 0, 2, 3, 11, 15, 16, 21, 43	
W	$-720x^3 + 1048x^2 - 2344x + 1200$
A	$-2x^5y^3 + 2x^5y + x^4y^4 + 9x^4y^3 - x^4y^2 - 9x^4y - 8x^3y^4 - 10x^3y^3 + 8x^3y^2 + 10x^3y + 2x^2y^5 + 17x^2y^4 - 7x^2y^3 - 17x^2y^2 + 5x^2y - 518400x^2 + 2xy^6 - 6xy^5 + 10xy^4 + 18xy^3 - 12xy^2 - 12xy + 1509120x - y^7 - y^6 - 62y^5 + 950y^4 - 11205y^3 + 137739y^2 + 379140y - 650560$
B	$4x^3y^6 - 28x^3y^5 + 514x^3y^4 - 5602x^3y^3 + 68826x^3y^2 + 189566x^3y + 1911552x^3 - 6x^2y^7 + 9x^2y^6 - 303x^2y^5 + 3024x^2y^4 - 37182x^2y^3 - 60329x^2y^2 - 860989x^2y - 2623936x^2 + 4xy^8 + 2xy^7 + 165xy^6 - 1605xy^5 + 20082xy^4 - 247696xy^3 + 400341xy^2 - 1192125xy + 3188160x - y^9 - 2y^8 - 71y^7 + 870y^6 - 10837y^5 + 134438y^4 + 424895y^3 + 865046y^2 + 4035902y - 789440$
$g = 2, i = 10, m = 1$, neighbours 27, 0, 4, 37, 13, 15	
W	$2x^2 - 2x - 12$
A	$-2x^2y + xy^2 + 4xy + y^3 - 4y$
B	$3x^3y^2 - 2x^3y + 4x^3 - 2x^2y^3 - 4x^2y^2 - 4x^2y - 16x^2 + 5xy^3 + 26xy - 12x + y^5 - y^4 + 2y^3 + 2y^2 - 38y + 72$
$g = 2, i = 11, m = 5$, neighbours 45, 2, 3, 31, 9, 36, 40, 16, 23, 44	
W	$-55608x^3 + 84192x^2 - 137928x + 46272$
A	$300x^7y^2 - 300x^7 + 300x^6y^3 - 300x^6y - 75x^5y^4 + 450x^5y^3 - 450x^5y + 75x^5 - 75x^4y^5 + 750x^4y^4 + 1200x^4y^3 + 150x^4y^2 - 1125x^4y - 900x^4 - 375x^3y^5 + 450x^3y^4 - 2400x^3y^3 - 1350x^3y^2 + 2775x^3y + 900x^3 - 450x^2y^6 - 1425x^2y^4 + 150x^2y^3 + 1650x^2y^2 - 150x^2y - 3092249439x^2 + 75xy^7 - 525xy^6 + 1725xy^5 - 1575xy^4 - 1275xy^3 + 2325xy^2 - 525xy + 9363497247x + 75y^8 + 4425y^6 - 97050y^5 + 1957725y^4 - 39231900y^3 + 786444975y^2 - 3395866794y - 3955965696$
B	$-375x^3y^7 - 2850x^3y^6 + 44475x^3y^5 - 984900x^3y^4 + 19616475x^3y^3 - 393216150x^3y^2 + 1697937297x^3y - 9236063508x^3 - 1950x^2y^7 + 24000x^2y^6 - 469800x^2y^5 + 9325500x^2y^4 - 186794550x^2y^3 + 652347936x^2y^2 - 3769068468x^2y + 11113620180x^2 + 225xy^9 - 900xy^8 + 13500xy^7 - 223425xy^6 + 4430250xy^5 - 88733775xy^4 + 1778896800xy^3 - 1558360491xy^2 + 16041274137xy - 10986612705x + 75y^{10} - 75y^9 + 5025y^8 - 102225y^7 + 2095425y^6 - 42098625y^5 + 844108575y^4 - 4551873519y^3 + 6848235348y^2 - 33228724500y - 1814867712$
$g = 2, i = 13, m = 1$, neighbours 0, 4, 33, 10, 14, 21	
W	$3x^4 + 8x^2 - x + 8$
A	$-9x^4 + 2x^3y - 3x^3 - x^2y^2 + 2x^2y - 48x^2 - 2xy^2 + 4xy + x + y^3 + 5y^2 + 28y - 41$
B	$-2x^5y + 3x^5 - 2x^3y^2 + 5x^3y + 11x^3 + x^2y^4 + 10x^2y^3 - 11x^2y^2$

	$+38x^2y - 59x^2 - 2xy^4 + 5xy^3 - 11xy^2 + 16xy + 56x + 4y^4 + 44y^3 + 48y^2 + 277y - 23$
$g = 2, i = 14, m = 1$, neighbours 1, 8, 13, 41	
W	$x^3 + x^2 - 2x$
A	$x^3 - 3x^2 - xy - 4x + 4y$
B	$-x^5 + x^4 - x^3y + x^3 + x^2y^2 + 2x^2y + 3x^2 - xy^3 - 8xy + 3x + 3y^3 - y^2 + 34y + 4$
$g = 2, i = 15, m = 1$, neighbours 29, 3, 9, 10, 23	
W	$-2x^4 + 6x^3 + 2x^2 - 6x$
A	$-4x^4 + 8x^3 - 28x^2 + 24x + 4y^2 - 32y + 28$
B	$16x^5 + 16x^4 - 16x^3y - 36x^3 + 4x^2y^3 - 4x^2y^2 + 28x^2y + 20x^2 - 4xy^3 + 6xy^2 - 20xy + 38x - 4y^4 + 30y^3 - 28y^2 + 218y + 24$
$g = 2, i = 16, m = 1$, neighbours 27, 3, 31, 8, 9, 11, 36, 42	
W	$-x^5 + 4x^4 - x^3 + 14x^2 - 12x + 8$
A	$-x^6 + 8x^5 - 18x^4 - 24x^3y^2 + 26x^3y - 45x^3 + x^2y^3 - 16x^2y - 120x^2 - 24xy^3 + 20xy^2 - 80xy + 103x + y^4 - 21y^3 + 3y^2 - 166y - 183$
B	$24x^5y^2 - 10x^5y + 66x^5 + 24x^4y^2 - 10x^4y + 66x^4 - 24x^3y^3 + 10x^3y^2 - 66x^3y + 59x^3 + 3x^2y^4 + 25x^2y^3 + 19x^2y^2 + 152x^2y - 75x^2 + 27xy^5 - 27xy^4 + 255xy^3 - 103xy^2 + 542xy + 389x - 2y^6 + 23y^5 - 31y^4 + 364y^3 - 187y^2 + 1512y + 119$
$g = 2, i = 18, m = 1$, neighbours 1, 3, 35, 37, 23	
W	$x^5 - x^4 + 3x^3 + x^2 - 4$
A	$-x^6 + 2x^5 - 7x^4 - 3x^3y^2 + 8x^3y - 43x^3 + x^2y^3 + 8x^2y - 8x^2 - 2xy^3 + 3xy^2 - 22xy - 6x + y^4 + 3y^3 + 13y^2 + 49y + 40$
B	$3x^5y^2 - 8x^5y + 48x^5 - 3x^4y^2 + 8x^4y - 48x^4 - 3x^3y^3 + 5x^3y^2 - 40x^3y + 11x^3 - x^2y^4 + 10x^2y^3 - 15x^2y^2 + 70x^2y + 42x^2 + 3xy^5 - 4xy^4 + 58xy^3 - 22xy^2 + 290xy + 26x - y^6 - 2y^5 - 20y^4 - 50y^3 - 100y^2 - 309y - 56$
$g = 2, i = 21, m = 1$, neighbours 31, 9, 13, 41	
W	$x^3 + x^2 - 2x$
A	$-3x^3 + x^2 + 3xy + 5x + 2y + 2$
B	$3x^5 + 3x^4 + 3x^3y + x^2y^2 + 4x^2 + 3xy^3 - 2xy^2 + 22xy + 9x + y^3 + 19y$
$g = 2, i = 23, m = 1$, neighbours 34, 11, 15, 18	
W	$-2x^4 + 2x^2$
A	$-4x^4 - 5x^3 + 10x^2 - 3xy + 8x + 4y^2 - 3y - 7$
B	$5x^5 - 5x^4 + 5x^3y + 3x^3 + 4x^2y^3 - 3x^2y^2 + 31x^2y + 3x^2 + xy^3 + 2xy^2 + 14xy - 6x + 4y^4 - 8y^3 + 31y^2 - 51y - 6$
$g = 2, i = 31, m = 1$, neighbours 11, 36, 16, 21, 42	
W	$-3x^3 + 4x^2 + 28x + 16$
A	$4x^4 - 4x^3y - 16x^3 + x^2y^2 + 20x^2y + 14x^2 - 8xy^2 - 32xy + 8x - 8y^3 - 23y^2 - 62y - 15$
B	$-5x^3y^3 - 33x^3y^2 - 45x^3y - 173x^3 + x^2y^4 + 20x^2y^3 + 39x^2y^2 + 88x^2y + 287x^2 - 24xy^3 + 53xy^2 - 256xy + 651x - 8y^5 - 14y^4 - 109y^3 - 96y^2 - 425y + 240$
$g = 2, i = 32, m = 1$, neighbours 2	
W	-1
A	0
B	1
$g = 2, i = 33, m = 1$, neighbours 5, 37, 13, 43	
W	$2x^3 - 2x$
A	$4xy - 8x + y^2 - 6y + 8$
B	$x^3y + x^2y + 4xy^3 + 3xy^2 + 26xy + 4x + y^4 - 3y^3 + 5y^2 - 29y - 4$
$g = 2, i = 34, m = 1$, neighbours 23	
W	$-x$
A	0
B	1
$g = 2, i = 35, m = 1$, neighbours 37, 40, 18	
W	$-x^2 + 5$
A	$-2x - y + 1$
B	$2x^3 + 3x^2y + 4x^2 - xy^2 + xy - 6x - y^3 - 9y - 13$

$g = 2, i = 36, m = 1$, neighbours 0, 31, 11, 38, 16, 43	
W	$6x^3 + 10x^2 + 10x + 4$
A	$4x^3y - 10x^2y - 36x^2 - 3xy^3 + xy^2 - 120x + y^4 + 2y^3 + 3y^2 + 42y - 52$
B	$4x^3y^3 + 2x^3y^2 - 16x^3y - 84x^3 - 4x^2y^4 - 10x^2y^3 - 34x^2y^2$ $- 38x^2y - 64x^2 - xy^5 + 9xy^4 + 17xy^3 + 41xy^2 + 136xy + 20x + y^6 + 2y^5$ $+ 10y^4 + 70y^3 + 55y^2 + 406y + 36$
$g = 2, i = 37, m = 1$, neighbours 29, 1, 33, 10, 35, 38, 39, 18	
W	$16x^3 - 72x^2 + 80x$
A	$8x^3y - 16x^3 + 16x^2y^2 - 24x^2y - 272x^2 + 2xy^3 + 20xy^2 - 80xy + 2368x$ $+ 2y^4 - 10y^3 + 24y^2 + 1112y - 2272$
B	$6x^3y^4 - 24x^3y^3 + 34x^3y^2 + 572x^3y + 2688x^3 - 2x^2y^5$ $+ 24x^2y^4 - 58x^2y^3 - 178x^2y^2 - 904x^2y - 3224x^2 + 2xy^5 + 42xy^4$ $- 54xy^3 + 168xy^2 - 440xy + 1056x - 8y^5 + 44y^4 + 1106y^3 + 58y^2 + 9068y + 1136$
$g = 2, i = 38, m = 1$, neighbours 6, 36, 37	
W	$6x - 6$
A	0
B	$2x^2 + 3xy - 19x + y^2 - 6y + 17$
$g = 2, i = 39, m = 3$, neighbours 37, 44	
W	$2x^3 + 12x^2 + 36x + 40$
A	$-4x^2 - 48x + 16y - 96$
B	$-8x^3y - 96x^3 - 4x^2y^2 - 40x^2y - 423x^2 - 16xy^2 - 160xy - 661x + 16y^3$ $- 64y^2 + 9y - 734$
$g = 2, i = 40, m = 1$, neighbours 2, 4, 6, 7, 35, 11, 43	
W	$-x^5 - x^4 - 8x^3 - 5x^2 + 3x$
A	$-x^6 - 2x^5 - 17x^4 + 7x^3y^2 - 16x^3y + 12x^3 + x^2y^3 + x^2y^2$ $+ 16x^2y - 58x^2 + 6xy^3 - 9xy^2 + 2xy - 4x + y^4 - 4y^3 + 42y^2 - 57y$
B	$-7x^5y^2 + 14x^5y - 39x^5 + 7x^3y^3 - 14x^3y^2 + 39x^3y - 52x^3$ $- 3x^2y^4 + 14x^2y^3 - 30x^2y^2 + 218x^2y + 13x^2 - 9xy^5 + 23xy^4$ $- 67xy^3 + 64xy^2 - 26xy + 83x - 48y^4 + 96y^3 - 339y^2 + 410y + 61$
$g = 2, i = 41, m = 1$, neighbours 1, 14, 21	
W	$x^2 + x - 2$
A	0
B	$-x^3 - x^2y - x^2 + xy - 2x + y^2 - y + 2$
$g = 2, i = 42, m = 1$, neighbours 31, 16, 43	
W	$x^2 - 3x$
A	0
B	$-x^3 - x^2y - 2x^2 - xy + 5x - y^2 + y + 2$
$g = 2, i = 43, m = 1$, neighbours 27, 33, 9, 36, 40, 42	
W	$x^5 - 3x^4 - x^3 + 6x^2 - 2x - 10$
A	$-x^6 + 6x^5 - 7x^4 - 5x^3y^2 - 12x^3y - 85x^3 + x^2y^3 + 4x^2y + 42x^2$ $- 6xy^3 - 40xy + 202x + y^4 - 8y^3 + 7y^2 - 36y + 88$
B	$5x^5y^2 + 12x^5y + 68x^5 + 5x^4y^2 + 12x^4y + 68x^4 - 5x^3y^3$ $+ 3x^3y^2 - 32x^3y + 7x^3 - x^2y^4 + 6x^2y^3 - 13x^2y^2 + 45x^2y$ $+ 53x^2 + 5xy^5 + 16xy^4 + 115xy^3 + 142xy^2 + 636xy + 118x - y^6 + 6y^5 - 2y^4$ $+ 92y^3 + 49y^2 + 382y + 94$
$g = 2, i = 44, m = 1$, neighbours 29, 2, 7, 11, 39	
W	$-x^4 - 2x^3 - 3x^2 - 2x - 1$
A	$-x^4 + 2x^3 - 10x^2 + 7x + y^2 - 4y - 13$
B	$-6x^5 - 6x^4 + 6x^3y - 16x^3 + x^2y^3 - x^2y^2 + 24x^2y - 11x^2 - xy^3$ $- 7xy^2 + 4xy + 5x - y^4 + 8y^3 - 13y^2 + 51y + 8$

Table 2: The data of the identities (3).

$\{i, j\} = \{0, 3\}, k_{ij} = 1$	
α	0
β	-1
γ	1
$\{i, j\} = \{0, 4\}, k_{ij} = 1$	

α	1
β	$x^3 + y^2 + 9$
γ	$-x^3 - 1$
$\{i, j\}$	$\{0, 11\}, k_{ij} = 5^2$
α	16
β	$8x^3 + 4x^2 + 2x + 16y^2 + 137$
γ	$-8x^3 - 4x^2 - 2x - 9$
$\{i, j\}$	$\{0, 13\}, k_{ij} = 5$
α	$x^3 + 2$
β	$x^5 + x^3y^2 + 7x^3 + 3x^2 + 2x + 2y^2 + 13$
γ	$-x^5 + x^3 - 3x^2 - 2x + 3$
$\{i, j\}$	$\{0, 14\}, k_{ij} = 1$
α	1
β	$x^2 - x + y^2 + 9$
γ	$-x^2 + x - 1$
$\{i, j\}$	$\{0, 16\}, k_{ij} = 1$
α	1
β	$-x^2 - x + y^2 + 8$
γ	$-x^2 - x$
$\{i, j\}$	$\{0, 25\}, k_{ij} = 3^3 \cdot 5$
α	$4x^3 + 41$
β	$-4x^5 + 4x^4 + 4x^3y^2 + 20x^3 - 25x^2 - 3x + 41y^2 + 281$
γ	$-4x^5 + 4x^4 - 12x^3 - 25x^2 - 3x - 47$
$\{i, j\}$	$\{1, 5\}, k_{ij} = 1$
α	1
β	$x^3 - x^2y + xy^2 + y^3 + 2y^2 + 15$
γ	$-y^3 - y^2 - 7$
$\{i, j\}$	$\{1, 9\}, k_{ij} = 2 \cdot 3$
α	-1
β	$-x^3 + x^2y - xy^2 - y^3 - 15$
γ	$y^3 + 7$
$\{i, j\}$	$\{1, 20\}, k_{ij} = 2^4 \cdot 3$
α	$-7x^4 + 49y + 48$
β	$-7x^7 + 7x^6y - 14x^4y^2 - 7x^4 + 41x^3 - 41x^2y - 49xy^2 + 42xy - 385x + 49y^3$
γ	$+138y^2 + 350y + 818$
γ	$7x^4y^2 - 90y^2 + 42y - 385$
$\{i, j\}$	$\{2, 20\}, k_{ij} = 2^2 \cdot 3^3$
α	$-5x^3 - 13$
β	$-5x^5 - 10x^4 - 5x^3y^2 + 5x^3y - 60x^3 - 38x^2 - 21x - 13y^2 + 13y - 121$
γ	$5x^5 + 10x^4 + 15x^3 + 38x^2 + 21x + 4$
$\{i, j\}$	$\{3, 8\}, k_{ij} = 13^2$
α	16
β	$8x^3 + 4x^2 + 2x + 16y^2 + 16y + 153$
γ	$-8x^3 - 4x^2 - 2x - 9$
$\{i, j\}$	$\{3, 16\}, k_{ij} = 2^6$
α	$-x^3 + 3$
β	$x^5 + x^4 - x^3y^2 - x^3y - 10x^3 - 5x^2 + 7x + 3y^2 + 3y + 44$
γ	$x^5 + x^4 - x^3 - 5x^2 + 7x + 17$
$\{i, j\}$	$\{3, 18\}, k_{ij} = 2^5$
α	x^3
β	$-x^5 + x^4 + x^3y^2 + x^3y + 10x^3 - 4x^2 - 8x + 16$
γ	$-x^5 + x^4 + x^3 - 4x^2 - 8x + 16$
$\{i, j\}$	$\{4, 13\}, k_{ij} = 2^3 \cdot 3^2$
α	$x^4 - 8y + 9$
β	$-x^7 - x^6y + 2x^5y + x^5 + 2x^4y^2 + x^4y - x^4 - 9x^3 - 9x^2y - 8xy^2 + 10xy$
γ	$-63x - 8y^3 + 10y^2 - 55y - 9$
γ	$-x^4y^2 - 2x^4y - x^4 - y^2 - 10y + 63$
$\{i, j\}$	$\{5, 8\}, k_{ij} = 2^5$
α	32

β	$16x^3 - 8x^2y + 4xy^2 - y^3 + 16y^2 + 136$
γ	$y^3 + 16y^2 + 120$
$\{i, j\}$	$\{6, 18\}, k_{ij} = 2^6 \cdot 3$
α	$x^4 - 8y + 12$
β	$x^7 - x^6y + x^6 - x^5y + 12x^3 - 12x^2y + 12x^2 + 8xy^2 + 4xy + 64x - 8y^3 - 40y$
γ	$+128$ $-x^4y^2 + x^4y - x^4 - 4y^2 + 28y + 52$
$\{i, j\}$	$\{6, 25\}, k_{ij} = 3 \cdot 5^2$
α	$x^4 - 11y + 24$
β	$x^7 - x^6y + x^6 - x^5y - 3x^5 + 3x^4y - 6x^4 + 27x^3 - 27x^2y + 27x^2$
γ	$+11xy^2 - 2xy + 46x - 11y^3 - 3y^2 - 7y + 89$ $-x^4y^2 + x^4y - 4x^4 - 16y^2 + 52y + 19$
$\{i, j\}$	$\{7, 10\}, k_{ij} = 7$
α	16
β	$-16x^3 - 16x^2y + 16x^2 - 16xy^2 + 32xy - 16x - 8y^3 + 20y^2 + 14y + 9$
γ	$8y^3 - 28y^2 + 62y + 9$
$\{i, j\}$	$\{7, 13\}, k_{ij} = 2^5 \cdot 3^3 \cdot 5$
α	$17x^4 - 289y + 721$
β	$-17x^7 - 17x^6y + 17x^6 + 34x^5y + 51x^5 + 34x^4y^2 + 17x^4y - 68x^4$
γ	$-602x^3 - 602x^2y + 602x^2 - 289xy^2 + 456xy - 2461x - 289y^3 + 745y^2 - 2628y + 1740$ $-17x^4y^2 - 68x^4 - 313y^2 + 748y + 1859$
$\{i, j\}$	$\{7, 14\}, k_{ij} = 3^3$
α	$x^4 - 13y + 26$
β	$-x^7 - x^6y + x^6 + x^5y + 3x^5 + 2x^4y^2 + x^4y - 27x^3 - 27x^2y + 27x^2$
γ	$-13xy^2 - 90x - 13y^3 + 27y^2 - 131y + 1$ $-x^4y^2 + x^4y - 4x^4 - 14y^2 + 54y + 63$
$\{i, j\}$	$\{7, 18\}, k_{ij} = 2^4 \cdot 5$
α	$x^4 - 8y$
β	$-x^7 - x^6y + x^6 - x^5y + 4x^5 + 2x^4y + 4x^3 + 4x^2y - 4x^2 - 8xy^2 + 8xy - 80x$
γ	$-8y^3 + 4y^2 - 64y - 4$ $-x^4y^2 + 3x^4y - 5x^4 + 12y^2 - 16y + 84$
$\{i, j\}$	$\{7, 22\}, k_{ij} = 2 \cdot 5 \cdot 7$
α	$-6x^4 + 45y - 38$
β	$-24x^7 - 9x^6y + 9x^6 + 27x^5y + 54x^5 + 3x^4y^2 + 93x^4y - 141x^4 - 47x^3$
γ	$-552x^2y + 552x^2 - 270xy^2 + 756xy - 653x - 60y^3 + 574y^2 - 766y + 512$ $30x^6 - 15x^5y + 45x^5 - 9x^4y^2 + 6x^4y - 48x^4 + 105x^2y^2 - 285x^2y$ $+280x^2 + 105xy^2 + 220xy - 225x + 105y^3 - 567y^2 + 1133y - 404$
$\{i, j\}$	$\{7, 23\}, k_{ij} = 2^3$
α	$x^4 - 14y + 30$
β	$-x^7 - x^6y + x^6 + x^5y + 4x^5 + 2x^4y^2 + 2x^4y - 36x^3 - 36x^2y + 36x^2$
γ	$-14xy^2 + 2xy - 64x - 14y^3 + 38y^2 - 122y + 6$ $-x^4y^2 + x^4y - 5x^4 - 22y^2 + 70y + 28$
$\{i, j\}$	$\{9, 12\}, k_{ij} = 2^7 \cdot 5$
α	$21y + 43$
β	$42x^3y + 86x^3 - 441x + 21y^3 + y^2 + 166y - 156$
γ	$63x^2y + 129x^2 + 42xy^2 + 44xy - 86x - 63y^2 - 66y - 753$
$\{i, j\}$	$\{9, 15\}, k_{ij} = 2^5 \cdot 5$
α	$3y + 13$
β	$6x^3y + 26x^3 - 90x + 3y^3 + 13y^2 + 39y + 19$
γ	$9x^2y + 39x^2 + 6xy^2 + 29xy + 13x - 3y^2 - y - 128$
$\{i, j\}$	$\{9, 21\}, k_{ij} = 2^2 \cdot 3^3 \cdot 5$
α	$-3069x^4 + 7688y + 10667$
β	$5518x^7 + 62x^6y - 62x^6 + 558x^5y + 2294x^5 - 3131x^4y^2 - 18290x^4y$
γ	$-18228x^4 - 78984x^3 + 9826x^2y - 9826x^2 - 15376xy^2 + 14530xy + 120991x$ $+2046y^3 - 48170y^2 - 1629y + 60773$ $14105x^6 + 5642x^5y + 8463x^5 + 62x^4y^2 + 20739x^4y + 27218x^4$ $+5642x^2y^2 + 64883x^2y - 120120x^2 + 5642xy^2 + 5551xy - 60788x + 5642y^3$ $+55241y^2 - 40278y + 70900$
$\{i, j\}$	$\{9, 23\}, k_{ij} = 2^6$

α	$-2001x^4 + 39008y + 5096$
β	$3772x^7 + 23x^6y - 23x^6 + 253x^5y + 1909x^5 - 2024x^4y^2 + 7038x^4y$ $+ 20010x^4 + 52792x^3 + 13776x^2y - 13776x^2 - 16928xy^2 + 13904xy - 107844x$ $+ 23736y^3 - 19812y^2 + 183924y + 91856$
γ	$9545x^6 + 3818x^5y - 13363x^5 + 23x^4y^2 + 6187x^4y + 26749x^4$ $+ 15272x^2y^2 + 22908x^2y + 109228x^2 - 15272xy^2 + 57436xy - 189572x + 15272y^3$ $+ 33372y^2 + 74956y + 96888$
$\{i, j\}$	$\{10, 11\}, k_{ij} = 2^3 \cdot 5^2 \cdot 7$
α	-81
β	$-81x^3 + 81x^2y + 162x^2 - 81xy^2 - 324xy - 324x + 27y^3 + 90y^2 + 336y - 479$
γ	$-27y^3 - 198y^2 - 318y - 523$
$\{i, j\}$	$\{10, 13\}, k_{ij} = 2 \cdot 5^2$
α	$6x^4 + 27y + 20$
β	$-12x^7 + 3x^6y + 6x^6 + 6x^5y - 9x^5 + 3x^4y^2 + 6x^4y + 6x^4 - 40x^3$ $- 2x^2y - 4x^2 - 18xy^2 + 20xy - 33x + 18y^3 + 58y^2 + 62y + 2$
γ	$-18x^6 - 9x^5y - 18x^5 + 3x^4y^2 + 21x^4 + 9x^2y^2 - 36x^2y - 75x^2 - 42xy$ $- 84x + 9y^3 - 65y^2 - 27y + 34$
$\{i, j\}$	$\{10, 15\}, k_{ij} = 2^6 \cdot 5$
α	$45x^4 + 891y + 806$
β	$45x^7 + 36x^6y + 72x^6 - 36x^5y - 108x^5 + 81x^4y^2 + 72x^4y - 99x^4$ $+ 4253x^3 - 6404x^2y - 12808x^2 - 648xy^2 + 9932xy + 5208x + 810y^3 - 6408y^2 + 1952y$ $+ 9196$
γ	$81x^5y + 162x^5 + 36x^4y^2 + 189x^4y + 198x^4 + 81x^2y^2 - 1215x^2y$ $+ 3402x^2 + 81xy^2 - 3366xy - 900x - 81y^3 - 5675y^2 - 23667y - 17906$
$\{i, j\}$	$\{10, 19\}, k_{ij} = 2^8 \cdot 3 \cdot 5$
α	$-7x^4 + 147y + 368$
β	$-7x^7 + 7x^6y + 14x^6 + 14x^5y + 14x^5 + 42x^4y + 49x^4 + 263x^3 - 263x^2y$ $- 526x^2 - 147xy^2 - 190xy - 2563x + 147y^3 - 336y^2 + 606y - 1736$
γ	$7x^4y^2 + 42x^4y + 42x^4 - 410y^2 - 1242y - 3615$
$\{i, j\}$	$\{11, 16\}, k_{ij} = 2^3 \cdot 5^2$
α	$21x^4 - 25$
β	$-42x^7 + 21x^4y^2 - 42x^4y - 42x^4 - 34x^3 - 157x - 84y^3 + 80y^2 - 81y + 55$
γ	$63x^6 - 42x^5y + 21x^5 - 105x^4y - 42x^4 + 84x^2y^2 - 105x^2y + 72x^2$ $+ 84xy^2 - 139xy + 38x - 84y^3 + 105y^2 - 106y + 280$
$\{i, j\}$	$\{11, 18\}, k_{ij} = 2^6 \cdot 5$
α	$2223x^4 + 30004y + 29964$
β	$-4524x^7 + 13x^6y + 13x^6 - 65x^5y + 637x^5 + 2236x^4y^2 + 9308x^4y$ $+ 3250x^4 - 100638x^3 - 5344x^2y - 5344x^2 + 5408xy^2 - 11968xy + 12010x + 16354y^3$ $+ 20200y^2 + 174220y + 90230$
γ	$6825x^6 - 4550x^5y - 11375x^5 + 13x^4y^2 - 2119x^4y + 3289x^4$ $+ 13650x^2y^2 - 36400x^2y + 169050x^2 - 13650xy^2 - 53550xy - 259000x - 13650y^3$ $+ 20240y^2 - 65852y + 60586$
$\{i, j\}$	$\{11, 22\}, k_{ij} = 2^2 \cdot 3 \cdot 5 \cdot 7$
α	$-435x^4 + 3248y - 2778$
β	$-1160x^7 - 29x^6y - 29x^6 + 319x^5y + 1421x^5 - 406x^4y^2 + 2726x^4y$ $- 1885x^4 - 9006x^3 - 1824x^2y - 1824x^2 - 6728xy^2 - 2904xy + 533x - 58y^3$ $- 7189y^2 + 20833y - 11446$
γ	$2755x^6 - 1102x^5y + 1653x^5 - 29x^4y^2 + 957x^4y - 3973x^4 + 3306x^2y^2$ $- 3857x^2y + 23313x^2 + 3306xy^2 - 9215xy + 17955x + 3306y^3 + 7775y^2 + 20115y - 8248$
$\{i, j\}$	$\{11, 23\}, k_{ij} = 2^3 \cdot 3$
α	$5x^4 - 3$
β	$10x^7 + 5x^4y^2 + 40x^4y + 45x^4 - 16x^3 + 72x - 10y^3 + 27y^2 + 72y + 71$
γ	$-25x^6 + 10x^5y + 35x^5 - 35x^4y - 60x^4 + 10x^2y^2 - 50x^2y + 15x^2$ $- 10xy^2 + 34xy - 31x + 10y^3 - 30y^2 - 69y - 98$
$\{i, j\}$	$\{12, 13\}, k_{ij} = 2^3 \cdot 5^2$
α	$-2x + 20$
β	$2x^3 - 20x^2 + xy^2 + 10x - 10y^2 - 99$
γ	$-xy^2 - 2xy - 8x + 10y^2 + 20y + 81$
$\{i, j\}$	$\{12, 16\}, k_{ij} = 2^7$
α	$y - 1$

$$\begin{aligned}
\beta & -x^7 + x^6 - 2x^5 + x^4y - x^4 - 11x^3 + x^2y^2 + x^2y + 9x^2 - 54x - y^3 + 13y \\
& -44 \\
\gamma & x^7 - 2x^4y + 2x^4 + 11x^3 + 32x - 21y + 85 \\
\{i, j\} & = \{12, 17\}, k_{ij} = 1 \\
\alpha & 0 \\
\beta & 1 \\
\gamma & -1 \\
\{i, j\} & = \{12, 26\}, k_{ij} = 2^4 \cdot 5 \\
\alpha & -2y + 10 \\
\beta & -x^7 - 2x^6 + 2x^4y - 2x^4 + 5x^3 - 2x^2y^2 + 6x^2y - 2x^2 - 34x + 2y^3 - 8y^2 \\
& + 7y - 65 \\
\gamma & x^7 - 5x^3 + 34x + y - 5 \\
\{i, j\} & = \{13, 14\}, k_{ij} = 2^2 \cdot 3^3 \\
\alpha & -y + 6 \\
\beta & -x^7 - x^6 + x^5 - 2x^4y + 2x^4 + 9x^3 - x^2y^2 + 4x^2y + x^2 + 44x - y^3 \\
& + 5y^2 + 5y + 102 \\
\gamma & x^7 + x^4y - x^4 - 9x^3 - 53x - 8y - 50 \\
\{i, j\} & = \{13, 21\}, k_{ij} = 2^4 \cdot 3^2 \cdot 5 \\
\alpha & -y + 9 \\
\beta & x^7 - x^6 - 2x^5 - 3x^4y + 3x^4 - 21x^3 - x^2y^2 + 7x^2y + 7x^2 - 122x - y^3 \\
& + 8y^2 + 43y + 434 \\
\gamma & -x^7 + 2x^4y - 2x^4 + 21x^3 + 164x - 43y - 277 \\
\{i, j\} & = \{13, 26\}, k_{ij} = 2^4 \cdot 5^2 \\
\alpha & 6x^4 - 30y + 70 \\
\beta & -3x^7 - 12x^6 - 6x^5y + 6x^5 + 3x^4y^2 - 18x^4y + 18x^4 + 10x^3 - 30x^2y^2 \\
& + 30x^2y - 80x^2 + 20xy + 30x - 30y^3 + 50y^2 - 155y + 255 \\
\gamma & 3x^7 - 3x^4y^2 + 6x^4y - 6x^4 - 10x^3 - 50x + 10y^2 + 45y - 75 \\
\{i, j\} & = \{14, 24\}, k_{ij} = 1 \\
\alpha & y + 1 \\
\beta & x^7 + x^6 - x^5 + 3x^4y + 2x^4 + 9x^3 + x^2y^2 + 8x^2 - xy^2 - 7x + 3y^3 \\
& + 2y^2 + 25y + 19 \\
\gamma & -x^7 - 2x^4y - x^4 - 9x^3 - x - 2y^3 - y^2 - 17y - 10 \\
\{i, j\} & = \{15, 22\}, k_{ij} = 2^3 \cdot 3 \cdot 5 \\
\alpha & -2x^3 + 12 \\
\beta & 2x^5 + x^3y^2 + 11x^3 - 8x^2 + 10x - 6y^2 - 2y - 52 \\
\gamma & 2x^4 - x^3y^2 - 2x^3y - 7x^3 + 2x + 6y^2 + 10y + 56 \\
\{i, j\} & = \{15, 23\}, k_{ij} = 2^5 \cdot 3 \\
\alpha & 10x^4 + 10y - 12 \\
\beta & 5x^7 - 10x^5y - 5x^4y^2 + 5x^4 - 61x^3 - 10x^2y^2 - 110x^2y - 120x^2 \\
& - 10xy^2 + 12xy - 54x + 56y^2 + 16y \\
\gamma & -5x^7 + 5x^4y^2 - 10x^4y - 5x^4 + 61x^3 - 66x + 10y^3 + 34y^2 + 6y - 108 \\
\{i, j\} & = \{18, 19\}, k_{ij} = 2^5 \cdot 3 \\
\alpha & -y - 5 \\
\beta & -x^7 - x^6 + 2x^5 + 4x^4y - 6x^3 - x^2y^2 - 2x^2y - x^2 + xy^2 + 2xy + 44x \\
& + 3y^3 + 7y^2 + 10y + 101 \\
\gamma & x^7 - 3x^4y - x^4 + 6x^3 - 37x - 2y^3 - 5y^2 - 9y - 70 \\
\{i, j\} & = \{19, 25\}, k_{ij} = 3 \cdot 5 \\
\alpha & -y + 1 \\
\beta & x^7 + 3x^6 + 3x^4y - 4x^4 + 27x^3 + 3x^2y^2 - 15x^2y - 204x^2 - 6xy^2 + 30xy \\
& - 301x + y^3 - 5y^2 - 225y + 482 \\
\gamma & -x^7 - 4x^6 - 4x^5 - 2x^4y - 27x^3 - 4x^2y^2 + 16x^2y + 176x^2 + 4xy^2 - 16xy \\
& + 479x + 4y^2 + 241y - 7 \\
\{i, j\} & = \{20, 21\}, k_{ij} = 2 \cdot 3 \\
\alpha & -y - 2 \\
\beta & -x^7 + x^6 + x^5 - 20x^3 + x^2y^2 + 4x^2y - 32x^2 + 2xy^2 + 8xy + 52x + y^3 \\
& - 2y^2 + 43y + 44 \\
\gamma & x^7 - 2x^6 - x^4y + 20x^3 - 2x^2y^2 - 4x^2y + 14x^2 - 2xy^2 - 4xy - 82x - 2y^3 \\
& + 2y^2 - 53y - 8 \\
\{i, j\} & = \{20, 24\}, k_{ij} = 2^2 \cdot 7
\end{aligned}$$

α	$4y + 28$
β	$x^7 + 4x^6 + 8x^5 + 8x^4 + 3x^3 + 4x^2y^2 + 12x^2y - 16x^2 + 8xy^2 + 24xy + 41x$
	$+ 8y^2 + 25y + 121$
γ	$-x^7 + 4x^4y + 8x^4 - 3x^3 - 73x + 4y^3 + 20y^2 + 7y + 107$
$\{i, j\} = \{25, 26\}, k_{ij} = 5^2$	
α	$-y + 3$
β	$-x^7 - 3x^6 + 2x^4 - 42x^3 - 3x^2y^2 + 18x^2y + 135x^2 + 3xy^2 - 18xy + 476x$
	$+ y^3 + 10y^2 + 216y - 232$
γ	$x^7 + 2x^6 - 2x^5 + x^4y - 2x^4 + 42x^3 + 2x^2y^2 - 16x^2y - 178x^2 - 4xy^2$
	$+ 32xy - 297x - 11y^2 - 199y + 442$
$\{i, j\} = \{0, 6\}, k_{ij} = 3$	
α	1
β	$-x^3 - x^2 - x + y^2 + 6$
γ	$x^3 + x^2 + x + 2$
$\{i, j\} = \{0, 9\}, k_{ij} = 5^2$	
α	16
β	$-8x^3 - 4x^2 - 2x + 16y^2 + 119$
γ	$8x^3 + 4x^2 + 2x + 9$
$\{i, j\} = \{0, 10\}, k_{ij} = 3 \cdot 5$	
α	1
β	$-x^3 + 2x^2 - 4x + y^2 + 15$
γ	$x^3 - 2x^2 + 4x - 7$
$\{i, j\} = \{0, 36\}, k_{ij} = 3^2$	
α	16
β	$8x^3 - 4x^2 + 2x + 16y^2 + 135$
γ	$-8x^3 + 4x^2 - 2x - 7$
$\{i, j\} = \{1, 7\}, k_{ij} = 3 \cdot 5^2$	
α	16
β	$16x^3 - 16x^2y + 16xy^2 - 8y^3 + 12y^2 + 6y + 75$
γ	$8y^3 + 12y^2 + 6y + 59$
$\{i, j\} = \{1, 14\}, k_{ij} = 3 \cdot 13$	
α	$38x^4 - 323y + 39$
β	$-76x^7 + 19x^6y + 57x^5y + 114x^5 + 19x^4y^2 - 228x^4y + 38x^4 - 1201x^3$
	$+ 448x^2y^2 - 361xy^2 + 603xy + 1253x - 95y^3 + 123y^2 - 2719y + 14$
γ	$-114x^6 - 57x^5y + 114x^5 + 19x^4y^2 - 228x^2y^2 + 57x^2y - 1278x^2$
	$+ 228xy^2 - 810xy + 1278x - 228y^3 - 84y^2 + 135y - 25$
$\{i, j\} = \{1, 18\}, k_{ij} = 2^3 \cdot 5$	
α	$x^4 - 6y + 1$
β	$x^7 - x^6y - x^5y + x^5 - 2x^4y + 2x^4 + 3x^3 - 3x^2y + 6xy^2 - xy + 39x - 6y^3$
	$+ 4y^2 - 46y + 40$
γ	$-x^4y^2 - x^4y + x^4 + 3y^2 - y + 39$
$\{i, j\} = \{1, 37\}, k_{ij} = 5$	
α	-1
β	$-x^3 + x^2y - xy^2 - y^3 + 2y^2 - 4y - 7$
γ	$y^3 - y^2 + 2y + 3$
$\{i, j\} = \{1, 41\}, k_{ij} = 2^2 \cdot 3 \cdot 13$	
α	$-x^4 + 15y - 10$
β	$-x^7 + x^6y + 2x^5y + 4x^5 - 2x^4y^2 - 4x^4y - x^4 - 19x^3 + 19x^2y - 15xy^2$
	$+ 14xy - 83x + 15y^3 - 14y^2 + 98y - 10$
γ	$x^4y^2 + 2x^4y + 4x^4 + 4y^2 + 14y - 83$
$\{i, j\} = \{2, 9\}, k_{ij} = 2^2 \cdot 3$	
α	-2
β	$x^3 + x^2 + x - 2y^2 + 2y - 16$
γ	$-x^3 - x^2 - x - 2$
$\{i, j\} = \{2, 11\}, k_{ij} = 2^4$	
α	-2
β	$-x^3 - 2y^2 + 2y - 19$
γ	$x^3 + 1$
$\{i, j\} = \{2, 32\}, k_{ij} = 1$	

α	0
β	1
γ	-1
$\{i, j\}$	$\{2, 40\}, k_{ij} = 2^4 \cdot 3$
α	$-x^3 - 7$
β	$x^5 - x^3y^2 + x^3y - 8x^3 + 8x^2 - 7x - 7y^2 + 7y - 55$
γ	$x^5 + x^3 + 8x^2 - 7x + 8$
$\{i, j\}$	$\{2, 44\}, k_{ij} = 2^3$
α	-1
β	$x^2 + x - y^2 + y - 9$
γ	$x^2 + x$
$\{i, j\}$	$\{3, 9\}, k_{ij} = 2^2 \cdot 5$
α	2
β	$-x^3 + 2y^2 + 2y + 17$
γ	$x^3 + 1$
$\{i, j\}$	$\{3, 11\}, k_{ij} = 2^3 \cdot 3$
α	2
β	$x^3 + x^2 + x + 2y^2 + 2y + 20$
γ	$-x^3 - x^2 - x - 2$
$\{i, j\}$	$\{3, 15\}, k_{ij} = 2^2 \cdot 3 \cdot 5$
α	$-x^3 + 6$
β	$x^5 + x^4 - x^3y^2 - x^3y - 8x^3 - 4x^2 + 6x + 6y^2 + 6y + 60$
γ	$x^5 + x^4 + x^3 - 4x^2 + 6x + 6$
$\{i, j\}$	$\{4, 10\}, k_{ij} = 2^4$
α	-2
β	$2x^3 + 2x^2y + 2xy^2 + y^3 - 7$
γ	$-y^3 - 9$
$\{i, j\}$	$\{4, 40\}, k_{ij} = 2^4 \cdot 5$
α	$-9x^4 + 81y + 80$
β	$9x^7 + 9x^6y + 9x^4 - 71x^3 - 71x^2y + 81xy^2 - 72xy + 801x + 81y^3 - 72y^2 + 720y - 80$
γ	$9x^4y^2 - 152y^2 + 72y - 801$
$\{i, j\}$	$\{5, 6\}, k_{ij} = 2^3 \cdot 5^2$
α	8
β	$4x^3 - 2x^2y + xy^2 - y^3 - 9y^2 - 18y - 92$
γ	$y^3 + 18y^2 + 36y + 192$
$\{i, j\}$	$\{5, 33\}, k_{ij} = 2^4 \cdot 5^2$
α	16
β	$8x^3 - 4x^2y + 2xy^2 - y^3 + 8$
γ	$y^3 + 18y^2 + 36y + 192$
$\{i, j\}$	$\{6, 38\}, k_{ij} = 3^2$
α	3
β	$3x^3 - 3x^2y + 3x^2 + 3xy^2 - 6xy + 3x - y^3 + 3y^2 + 3y + 12$
γ	$y^3 - 3y^2 + 6y + 3$
$\{i, j\}$	$\{6, 40\}, k_{ij} = 2^3$
α	$x^4 - 2y + 2$
β	$x^7 + x^4y^2 - x^4y + x^4 + 6x^3 + 4x + 6y^2 - 12y + 6$
γ	$x^5y - x^5 - 2x^2y^2 - 4x^2y + 2x^2 + 6xy - 6x + 2y^3 + 4y^2 - 2y + 4$
$\{i, j\}$	$\{7, 40\}, k_{ij} = 2 \cdot 3 \cdot 7$
α	$5x^4 - 10y + 9$
β	$-5x^7 + 5x^4y^2 - 5x^4y - 5x^4 - 39x^3 - 42x + 39y^2 - 71y + 33$
γ	$-5x^5y + 5x^5 - 10x^2y^2 - 30x^2y + 30x^2 - 39xy + 39x + 10y^3 + 30y^2 - 30y + 42$
$\{i, j\}$	$\{7, 44\}, k_{ij} = 7^3$
α	$x^4 - 9y + 10$
β	$-x^7 - x^6y + x^6 + x^5y - x^5 - x^4y - 11x^3 - 11x^2y + 11x^2 - 9xy^2 - 8xy - 94x - 9y^3 - y^2 - 47y + 167$
γ	$-x^4y^2 + x^4y - 2y^2 + 30y + 83$
$\{i, j\}$	$\{8, 14\}, k_{ij} = 2^2 \cdot 3^3 \cdot 5^2$
α	$104x^4 - 1352y + 2700$

$$\begin{aligned}
&\beta \quad -52x^7 - 26x^6y + 234x^5y + 468x^5 + 130x^4y^2 + 468x^4y + 416x^4 - 1402x^3 \\
&\quad - 701x^2y - 2704xy^2 + 693xy - 21598x - 1352y^3 - 2007y^2 - 9430y - 22948 \\
&\gamma \quad -26x^4y^2 - 468x^4y - 936x^4 + 4707y^2 - 1386y + 43196 \\
&\{i, j\} = \{8, 16\}, k_{ij} = 3^3 \\
&\alpha \quad -2x^4 - 1 \\
&\beta \quad 4x^7 - 2x^4y^2 + 6x^4y + 4x^4 - 2x^3 - 14x + 4y^3 - 7y^2 - 19y + 11 \\
&\gamma \quad -6x^6 + 4x^5y - 6x^5 + 10x^4y - 4x^2y^2 + 10x^2y + 7x^2 - 4xy^2 + 8xy + 7x \\
&\quad + 4y^3 - 10y^2 - 9y + 28 \\
&\{i, j\} = \{9, 11\}, k_{ij} = 2 \cdot 5^2 \\
&\alpha \quad 32 \\
&\beta \quad 16x^3 - 8x^2y + 8x^2 + 4xy^2 - 8xy + 4x - y^3 + 18y^2 + 134 \\
&\gamma \quad y^3 + 12y^2 + 6y + 116 \\
&\{i, j\} = \{9, 16\}, k_{ij} = 2^5 \cdot 3 \\
&\alpha \quad 507x^4 + 3796y + 3273 \\
&\beta \quad 1170x^7 + 26x^6y - 26x^6 + 130x^5y - 390x^5 + 533x^4y^2 + 1378x^4y + 1352x^4 \\
&\quad + 16900x^3 - 1494x^2y + 1494x^2 - 2704xy^2 + 1890xy - 3651x + 2574y^3 + 3222y^2 \\
&\quad + 22223y + 19531 \\
&\gamma \quad 1833x^6 + 1222x^5y + 611x^5 + 26x^4y^2 - 403x^4y - 1976x^4 + 1222x^2y^2 \\
&\quad - 5499x^2y + 30644x^2 + 1222xy^2 + 8413xy + 16732x - 1222y^3 - 1403y^2 - 9964y - 22708 \\
&\{i, j\} = \{9, 43\}, k_{ij} = 2^5 \cdot 3^3 \cdot 5 \cdot 13 \\
&\alpha \quad -16x^4 + 256y + 2764 \\
&\beta \quad -8x^7 + 4x^6y - 4x^6 + 20x^5y - 36x^5 - 12x^4y^2 + 12x^4y + 8x^4 + 1446x^3 \\
&\quad - 723x^2y + 723x^2 - 512xy^2 - 1311xy - 11157x + 256y^3 + 1273y^2 + 5639y + 7322 \\
&\gamma \quad 4x^4y^2 + 32x^4y - 68x^4 - 1747y^2 - 1176y - 23037 \\
&\{i, j\} = \{10, 37\}, k_{ij} = 3 \\
&\alpha \quad -1 \\
&\beta \quad -x^3 + x^2y + 2x^2 - xy^2 - 4xy - 4x - y^3 + 1 \\
&\gamma \quad y^3 + 3y^2 + 6y + 3 \\
&\{i, j\} = \{11, 36\}, k_{ij} = 2 \\
&\alpha \quad 0 \\
&\beta \quad -1 \\
&\gamma \quad 1 \\
&\{i, j\} = \{11, 40\}, k_{ij} = 2^3 \cdot 7 \\
&\alpha \quad -150x^4 - 750y + 724 \\
&\beta \quad 300x^7 - 150x^4y^2 - 150x^4y + 75x^4 + 2302x^3 + 2060x - 1151y^2 + 254y + 668 \\
&\gamma \quad -450x^6 + 300x^5y + 300x^5 + 450x^4y - 750x^2y^2 + 1875x^2y - 5103x^2 + 2302xy \\
&\quad + 2302x + 750y^3 - 1875y^2 + 5103y - 4120 \\
&\{i, j\} = \{11, 44\}, k_{ij} = 2^4 \cdot 3 \cdot 5 \cdot 7^2 \\
&\alpha \quad 12x^4 - 216y - 496 \\
&\beta \quad -6x^7 - 3x^6y - 3x^6 + 21x^5y - 99x^5 + 9x^4y^2 - 63x^4y + 42x^4 + 68x^3 \\
&\quad + 34x^2y + 34x^2 - 432xy^2 + 770xy - 3558x - 216y^3 + 618y^2 - 2022y + 2044 \\
&\gamma \quad -3x^4y^2 - 48x^4y + 195x^4 + 898y^2 - 1472y + 7150 \\
&\{i, j\} = \{14, 41\}, k_{ij} = 3 \cdot 5 \cdot 13 \\
&\alpha \quad y + 3 \\
&\beta \quad -x^7 + x^6 + x^5 + x^4y - 21x^3 + x^2y^2 - 2x^2y + 16x^2 - xy^2 + 2xy - 47x \\
&\quad - y^3 + 2y^2 + 25y + 143 \\
&\gamma \quad x^7 - x^4 + 21x^3 + 73x + 2y^3 - 3y^2 - 11y - 96 \\
&\{i, j\} = \{16, 42\}, k_{ij} = 2^3 \\
&\alpha \quad -1 \\
&\beta \quad -x^5 + x^4 - 3x^2y + x^2 - 7x + 2y^2 - 4y + 17 \\
&\gamma \quad x^5 + x^3 + 2x^2y + x^2 + 9x - y^2 + 3y - 8 \\
&\{i, j\} = \{18, 23\}, k_{ij} = 2^4 \\
&\alpha \quad -2 \\
&\beta \quad 2x^2 + y^2 - 2y + 9 \\
&\gamma \quad -2x - y^2 - 9 \\
&\{i, j\} = \{21, 41\}, k_{ij} = 2^2 \cdot 3^2 \\
&\alpha \quad -y - 7 \\
&\beta \quad -x^7 - x^6 + 2x^3 - x^2y^2 - 2x^2y + 7x^2 - xy^2 - 2xy - 74x + y^3 + 3y^2 - 4y - 91 \\
&\gamma \quad x^7 - x^4y - x^4 - 2x^3 + 79x - 2y^3 - 7y^2 + 7y + 24
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{31, 11\}, k_{ij} = 2^3 \cdot 3 \\
&\alpha \quad -3 \\
&\beta \quad -3x^3 + 3x^2y + 3x^2 - 3xy^2 - 6xy - 3x + y^3 + y^2 + 5y - 18 \\
&\gamma \quad -y^3 - 4y^2 - 2y - 9 \\
&\{i, j\} = \{31, 16\}, k_{ij} = 2^8 \cdot 5 \\
&\alpha \quad 3x^4 - 36y - 196 \\
&\beta \quad 3x^7 - 3x^6y - 3x^6 + 3x^5y + 6x^5 - 6x^4y - 6x^4 - 184x^3 + 184x^2y + 184x^2 \\
&\quad + 36xy^2 - 316xy + 304x - 36y^3 + 60y^2 - 76y - 988 \\
&\gamma \quad -3x^4y^2 - 3x^4y + 3x^4 + 220y^2 + 52y + 488 \\
&\{i, j\} = \{31, 21\}, k_{ij} = 2^6 \\
&\alpha \quad x^4 - 10y - 18 \\
&\beta \quad x^7 - x^6y - x^6 - x^5y + 2x^5 + 2x^4y^2 + 2x^4y - 2x^4 - 16x^3 + 16x^2y \\
&\quad + 16x^2 + 10xy^2 - 2xy + 64x - 10y^3 - 34y^2 - 102y - 162 \\
&\gamma \quad -x^4y^2 - 3x^4y + x^4 + 26y^2 + 30y + 80 \\
&\{i, j\} = \{31, 36\}, k_{ij} = 2^4 \cdot 11 \\
&\alpha \quad -81 \\
&\beta \quad -81x^3 + 81x^2y + 81x^2 - 81xy^2 - 162xy - 81x + 27y^3 - 9y^2 + 3y - 514 \\
&\gamma \quad -27y^3 - 126y^2 - 120y - 257 \\
&\{i, j\} = \{31, 42\}, k_{ij} = 2^3 \cdot 11 \\
&\alpha \quad x - 11 \\
&\beta \quad x^4 - x^3y - 12x^3 + 13x^2y + 10x^2 - 23xy + 18x + 11y - 76 \\
&\gamma \quad -xy^2 - 2x + 11y^2 + 23 \\
&\{i, j\} = \{33, 13\}, k_{ij} = 3^3 \cdot 5^2 \\
&\alpha \quad -x^3 + 26 \\
&\beta \quad -x^5 - x^3y^2 - 2x^3y - 13x^3 + 25x^2 - 26x + 26y^2 + 52y + 337 \\
&\gamma \quad x^5 + x^3 - 25x^2 + 26x - 25 \\
&\{i, j\} = \{33, 37\}, k_{ij} = 2 \cdot 5^2 \\
&\alpha \quad 2 \\
&\beta \quad -x^3 + 2y^2 + 4y + 23 \\
&\gamma \quad x^3 + 1 \\
&\{i, j\} = \{33, 43\}, k_{ij} = 3^3 \cdot 5^2 \\
&\alpha \quad -2x^3 + 27 \\
&\beta \quad 2x^5 + 2x^4 - 2x^3y^2 - 4x^3y - 22x^3 - 23x^2 + 27x + 27y^2 + 54y + 351 \\
&\gamma \quad 2x^5 + 2x^4 + 2x^3 - 23x^2 + 27x + 27 \\
&\{i, j\} = \{34, 23\}, k_{ij} = 2^6 \\
&\alpha \quad x^4 - 12y - 20 \\
&\beta \quad -x^7 - x^6y - x^6 + x^5y - 2x^5 + 2x^4y^2 + 2x^4y - 4x^4 + 16x^3 + 16x^2y \\
&\quad + 16x^2 - 12xy^2 + 4xy - 104x - 12y^3 - 36y^2 - 116y - 204 \\
&\gamma \quad -x^4y^2 - 3x^4y + x^4 + 28y^2 + 28y + 120 \\
&\{i, j\} = \{35, 18\}, k_{ij} = 2^{10} \\
&\alpha \quad x^4 - 16y + 39 \\
&\beta \quad x^7 - x^6y + 2x^6 - x^5y - x^5 + 2x^4y - 6x^4 + 23x^3 - 23x^2y + 46x^2 \\
&\quad + 16xy^2 + 25xy + 169x - 16y^3 - 18y + 454 \\
&\gamma \quad -x^4y^2 + 3x^4y - 5x^4 - 7y^2 + 117y + 77 \\
&\{i, j\} = \{35, 37\}, k_{ij} = 5^2 \\
&\alpha \quad 1 \\
&\beta \quad x^3 - x^2y + 2x^2 + xy^2 - 4xy + 4x + y^3 - 4y^2 + 16y + 15 \\
&\gamma \quad -y^3 + 5y^2 - 14y - 3 \\
&\{i, j\} = \{35, 40\}, k_{ij} = 2^4 \cdot 11^2 \\
&\alpha \quad 5x^4 - 75y + 148 \\
&\beta \quad 5x^7 - 5x^6y + 10x^6 - 20x^5 + 20x^4y - 35x^4 + 83x^3 - 83x^2y + 166x^2 \\
&\quad + 75xy^2 + 160xy + 373x - 75y^3 - 10y^2 - 128y + 894 \\
&\gamma \quad -5x^4y^2 + 20x^4y - 40x^4 - 8y^2 + 492y + 41 \\
&\{i, j\} = \{36, 16\}, k_{ij} = 2^7 \\
&\alpha \quad 4x^4 - 12y \\
&\beta \quad -2x^7 - x^6y + x^6 + 7x^5y - x^5 + 3x^4y^2 - 10x^4y - x^4 - 2x^3 - x^2y \\
&\quad + x^2 - 24xy^2 - xy - 37x - 12y^3 + 35y^2 - 10y + 55 \\
&\gamma \quad -x^4y^2 - 12x^4y + x^4 + 47y^2 + 4y + 73 \\
&\{i, j\} = \{36, 38\}, k_{ij} = 3
\end{aligned}$$

α	0
β	1
γ	-1
$\{i, j\}$	$\{36, 43\}, k_{ij} = 2^5 \cdot 3^2 \cdot 11$
α	$-4x^4 + 48y - 4$
β	$2x^7 + x^6y - x^6 - 7x^5y - 5x^5 - 3x^4y^2 + 7x^4y + 10x^4 + 10x^3 + 5x^2y$ $- 5x^2 + 96xy^2 - 3xy + 1271x + 48y^3 - 143y^2 + 611y - 1902$
γ	$x^4y^2 + 12x^4y + 11x^4 - 187y^2 - 4y - 2537$
$\{i, j\}$	$\{37, 18\}, k_{ij} = 2^2 \cdot 3^4 \cdot 5$
α	$-16x^4 + 32y - 52$
β	$-8x^7 + 4x^6y - 8x^6 + 28x^5y + 56x^5 - 12x^4y^2 + 20x^4y + 112x^4 + 142x^3$ $- 71x^2y + 142x^2 - 64xy^2 + 47xy + 382x + 32y^3 - 187y^2 + 61y + 596$
γ	$4x^4y^2 + 40x^4y + 128x^4 - 199y^2 + 378y + 480$
$\{i, j\}$	$\{37, 38\}, k_{ij} = 2^5 \cdot 3$
α	32
β	$16x^3 - 8x^2y + 16x^2 + 4xy^2 - 16xy + 16x - y^3 + 20y^2 + 120$
γ	$y^3 + 8y^2 + 24y + 88$
$\{i, j\}$	$\{37, 39\}, k_{ij} = 2^5 \cdot 5^2$
α	32
β	$16x^3 - 8x^2y + 16x^2 + 4xy^2 - 16xy + 16x - y^3 + 22y^2 + 20y + 208$
γ	$y^3 + 10y^2 + 44y + 176$
$\{i, j\}$	$\{39, 44\}, k_{ij} = 2^2 \cdot 3^3$
α	-4
β	$2x^3 + x^2y - 2x^2 - 7xy - 12x - 3y^2 + 6y + 28$
γ	$y^2 + 10y + 28$
$\{i, j\}$	$\{40, 43\}, k_{ij} = 3^4$
α	$y + 3$
β	$-x^7 + x^6 + 2x^5 - 3x^4y - 11x^3 + x^2y^2 + 3x^2y + 8x^2 + x - y^3 - 3y^2 - 31y$ $+ 60$
γ	$x^7 + 2x^4y + 11x^3 + 21x + 23y - 39$
$\{i, j\}$	$\{42, 43\}, k_{ij} = 2 \cdot 3^2 \cdot 11$
α	$y - 14$
β	$x^7 + x^6 - x^5 + 2x^4 - 3x^3 + x^2y^2 - x^2 + 2xy^2 + 84x + 3y^3 - y^2 + 7y + 132$
γ	$-x^7 - x^4y - 5x^4 + 3x^3 - 77x - 4y^3 + 4y^2 - 6y - 40$

Table 3: The polynomials $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$ of Step 2 and the signs of H_i for $x \geq 9$ and for $x \leq -11$.

i	k_i	$\varphi_i(x)$	$x \geq 9$	$x \leq -11$
0	-1	$-x^4 - x - 1$	+	+
1	-1	$-x^4 + x^3 + 7x - 1$	+	+
2	-1	$-x^4 - x + 8$	+	+
3	-1	$-x^4 - x - 10$	+	+
4	-1	$-x^4 - x^3 - 9x - 1$	+	+
5	-1	$-x^4 + 8x^3 + 15x - 1$	+	+
6	-1	$-x^4 + x^3 - 3x^2 + 10x - 10$	+	+
7	-1	$-x^4 - x^3 - 3x^2 - 12x - 10$	+	+
8	-1	$-x^4 - 8x^3 - 17x - 1$	+	+
9	-1	$-x^4 + 8x^3 - 12x^2 + 21x - 10$	+	+
10	-1	$-x^4 + x^3 + 6x^2 + 19x + 23$	+	+
11	-1	$-x^4 - 8x^3 + 12x^2 - 23x + 8$	+	+
12	1	$-x^6 + 4x^4 - 11x^2 + x + 10$	+	+
13	-1	$-x^6 - 4x^4 - 11x^2 - x - 10$	+	+
14	-1	$-x^6 - 3x^5 - 4x^4 - x^3 - 8x^2 - 9x - 1$	+	+
15	1	$-x^6 + 3x^5 + x^4 - 5x^3 - 8x^2 + 12x + 10$	+	+
16	1	$-x^6 + 3x^5 - 5x^4 + 7x^3 - 14x^2 + 12x - 8$	+	+
17	1	$-x^6 + 7x^4 - 20x^2 + x + 25$	+	+
18	1	$-x^6 - 3x^5 - 5x^4 - 7x^3 - 14x^2 - 10x - 8$	+	+
19	1	$-x^6 - 6x^5 - 11x^4 - 8x^3 - 8x^2 - 15x + 1$	+	+
20	-1	$-x^6 + 6x^5 - 13x^4 + 8x^3 - 8x^2 + 15x - 1$	+	+

i	k_i	$\varphi_i(x)$	$x \geq 9$	$x \leq -11$
21	-1	$-x^6 + 3x^5 - x^4 - 5x^3 - 8x^2 + 10x + 8$	+	+
22	-1	$-x^6 + 3x^5 - 7x^4 + 7x^3 - 14x^2 + 10x - 10$	+	+
23	-1	$-x^6 - 3x^5 - x^4 + 5x^3 - 8x^2 - 12x + 8$	+	+
24	-1	$-x^6 - 6x^5 - 13x^4 - 8x^3 - 8x^2 - 17x - 1$	+	+
25	1	$-x^6 - 3x^5 + 4x^4 + 11x^3 - 14x^2 - 19x + 25$	+	+
26	1	$-x^6 - 6x^5 - 8x^4 + 4x^3 + x^2 - 21x + 10$	+	+
31	-1	$-x^4 + x^3 + 3x^2 + 10x + 8$	+	+
32	-1	$-x^4 - x + 23$	+	+
33	-1	$-x^4 - x - 25$	+	+
34	-1	$-x^4 - x^3 + 3x^2 - 12x + 8$	+	+
35	-1	$-x^4 + x^3 - 6x^2 + 19x - 25$	+	+
36	-1	$-x^4 - 8x^3 - 12x^2 - 23x - 10$	+	+
37	-1	$-x^4 + 8x^3 - 24x^2 + 39x - 25$	+	+
38	-1	$-x^4 - 8x^3 + 24x^2 - 41x + 23$	+	+
39	-1	$-x^4 - 8x^3 - 24x^2 - 41x - 25$	+	+
40	1	$-x^6 + x^4 - 8x^2 + x + 1$	+	+
41	-1	$-x^6 + 5x^4 - 20x^2 - x + 23$	+	+
42	1	$-x^6 + 6x^5 - 14x^4 + 20x^3 - 23x^2 + 23x - 8$	+	+
43	1	$-x^6 + 3x^5 + 4x^4 - 11x^3 - 14x^2 + 21x + 25$	+	+
44	1	$-x^6 + 3x^5 - 8x^4 + 13x^3 - 26x^2 + 21x - 23$	+	+

Table 4: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 4, grouped by p, k and the value of $(B_p^{(g)})_g$.

p	k	value	classes (a, b)
2	6	(1, 1)	(0, 23), (15, 39), (16, 39), (31, 55), (32, 55), (46, 41), (47, 7), (48, 7), (63, 23)
2	7	(1, 1)	(1, 13), (4, 91), (7, 95), (8, 127), (9, 21), (12, 67), (17, 29), (20, 43), (23, 111), (24, 79), (25, 37), (26, 21), (28, 19), (33, 45), (36, 123), (39, 127), (40, 31), (41, 53), (44, 99), (45, 105), (49, 61), (52, 75), (55, 15), (56, 111), (57, 69), (60, 51), (65, 77), (68, 27), (70, 97), (71, 31), (72, 63), (73, 85), (76, 3), (81, 93), (84, 107), (87, 47), (88, 15), (89, 101), (90, 85), (92, 83), (97, 109), (100, 59), (103, 63), (104, 95), (105, 117), (108, 35), (113, 125), (116, 11), (119, 79), (120, 47), (121, 5), (124, 115)
2	8	(1, 1)	(2, 93), (3, 107), (5, 129), (6, 161), (10, 197), (11, 51), (18, 173), (21, 145), (27, 67), (35, 139), (37, 161), (42, 101), (43, 83), (50, 77), (53, 177), (58, 181), (59, 99), (66, 157), (67, 171), (69, 193), (74, 5), (75, 115), (82, 237), (85, 209), (91, 131), (99, 203), (101, 225), (102, 129), (106, 165), (107, 147), (114, 141), (117, 241), (122, 245), (123, 163), (125, 57), (130, 221), (131, 235), (133, 1), (134, 33), (138, 69), (139, 179), (146, 45), (147, 251), (149, 17), (155, 195), (163, 11), (165, 33), (170, 229), (171, 211), (178, 205), (179, 27), (181, 49), (186, 53), (187, 227), (194, 29), (195, 43), (197, 65), (202, 133), (203, 243), (210, 109), (213, 81), (219, 3), (227, 75), (229, 97), (234, 37), (235, 19), (242, 13), (245, 113), (250, 117), (251, 35), (253, 185)

p	k	value	classes (a, b)
2	9	(1, 1)	(13, 457), (14, 73), (19, 123), (22, 369), (29, 473), (30, 25), (34, 509), (38, 65), (51, 155), (54, 273), (61, 505), (62, 441), (77, 9), (78, 393), (83, 187), (86, 177), (93, 25), (94, 345), (98, 317), (109, 41), (115, 219), (118, 81), (126, 249), (141, 73), (142, 201), (150, 497), (157, 89), (158, 153), (162, 125), (166, 193), (182, 401), (189, 121), (190, 57), (205, 137), (206, 9), (211, 315), (214, 305), (221, 153), (222, 473), (226, 445), (230, 1), (237, 169), (243, 347), (246, 209), (254, 377), (269, 201), (270, 329), (275, 379), (278, 113), (285, 217), (286, 281), (290, 253), (294, 321), (307, 411), (310, 17), (317, 249), (318, 185), (333, 265), (334, 137), (339, 443), (342, 433), (349, 281), (350, 89), (354, 61), (365, 297), (371, 475), (374, 337), (382, 505), (397, 329), (398, 457), (406, 241), (413, 345), (414, 409), (418, 381), (422, 449), (438, 145), (445, 377), (446, 313), (461, 393), (462, 265), (467, 59), (470, 49), (477, 409), (478, 217), (482, 189), (486, 257), (493, 425), (499, 91), (502, 465), (510, 121)
3	2	(-1, 1)	(1, 3)
3	3	(-1, 1)	(4, 18), (7, 6), (13, 9), (22, 0), (25, 15)
3	3	(-1, -1)	(1, 22), (4, 1), (7, 7), (10, 13), (13, 19), (16, 25), (19, 4), (22, 10), (25, 16)
3	4	(1, -1)	(1, 29), (4, 35), (10, 47), (13, 53), (19, 65), (22, 71), (28, 2), (31, 8), (37, 20), (40, 26), (46, 38), (49, 44), (55, 56), (58, 62), (64, 74), (67, 80), (73, 11), (76, 17)
3	4	(-1, 1)	(43, 24), (70, 78)
3	5	(1, -1)	(7, 176), (16, 194), (25, 212), (34, 230), (43, 5), (52, 23), (61, 41), (70, 59), (79, 77), (88, 95), (97, 113), (106, 131), (115, 149), (124, 167), (133, 185), (142, 203), (151, 221), (160, 239), (169, 14), (178, 32), (187, 50), (196, 68), (205, 86), (214, 104), (223, 122), (232, 140), (241, 158)
3	5	(-1, 1)	(16, 132), (97, 51), (178, 213)
5	1	(1, 1)	(2, 4), (4, 1)
5	2	(1, 1)	(0, 7), (0, 12), (0, 16), (0, 17), (0, 22), (5, 11), (10, 6), (12, 3), (12, 8), (12, 18), (12, 23), (20, 21), (24, 7), (24, 12), (24, 17)
5	3	(1, 1)	(8, 65), (13, 100), (15, 101), (18, 110), (23, 95), (24, 22), (24, 47), (24, 72), (24, 97), (24, 122), (33, 115), (37, 13), (37, 38), (37, 63), (37, 88), (37, 113), (38, 25), (40, 76), (43, 35), (48, 20), (49, 2), (49, 27), (49, 52), (49, 77), (49, 102), (58, 40), (63, 75), (65, 51), (68, 85), (73, 70), (83, 90), (88, 0), (90, 26), (93, 10), (98, 120), (100, 2), (100, 27), (100, 52), (100, 77), (100, 102), (108, 15), (113, 50), (115, 1), (118, 60), (123, 45)
5	4	(1, 1)	(3, 130), (28, 180), (53, 230), (78, 280), (103, 330), (128, 380), (153, 430), (178, 480), (203, 530), (228, 580), (253, 5), (278, 55), (303, 105), (328, 155), (353, 205), (378, 255), (403, 305), (428, 355), (453, 405), (478, 455), (503, 505), (528, 555), (553, 605), (578, 30), (603, 80)
7	1	(1, 1)	(1, 5), (2, 1), (4, 4), (4, 6)
7	2	(1, 1)	(37, 3), (37, 10), (37, 17), (37, 24), (37, 31), (37, 45)
7	3	(1, 1)	(37, 87), (37, 136), (37, 185), (37, 234), (37, 283), (37, 332)
7	4	(1, 1)	(723, 38), (723, 381), (723, 724), (723, 1067), (723, 1410), (723, 1753), (723, 2096)
11	1	(1, 1)	(1, 4), (2, 7), (4, 7), (5, 3), (7, 0), (7, 5), (8, 1), (8, 9), (9, 3)
11	2	(1, 1)	(3, 7), (7, 105), (14, 18), (18, 28), (25, 29), (29, 72), (36, 40), (40, 116), (47, 51), (51, 39), (58, 62), (62, 83), (69, 73), (73, 6), (80, 84), (84, 50), (91, 95), (95, 94), (102, 106), (113, 117), (117, 61)
11	3	(1, 1)	(106, 985), (227, 138), (348, 622), (469, 1106), (590, 259), (711, 743), (832, 1227), (953, 380), (1074, 864), (1195, 17), (1316, 501)
13	1	(1, 1)	(0, 3), (4, 3), (7, 5), (7, 7), (10, 3), (12, 3)

p	k	value	classes (a, b)
13	2	(1, 1)	(7, 27), (11, 54), (20, 79), (24, 15), (33, 131), (37, 145), (46, 14), (50, 106), (59, 66), (63, 67), (72, 118), (76, 28), (89, 158), (98, 53), (102, 119), (111, 105), (115, 80), (124, 157), (128, 41), (137, 40), (141, 2), (150, 92), (154, 132), (163, 144), (167, 93)
13	3	(1, 1)	(85, 1), (254, 677), (423, 1353), (592, 2029), (761, 508), (930, 1184), (1099, 1860), (1268, 339), (1437, 1015), (1606, 1691), (1775, 170), (1944, 846), (2113, 1522)